

When Do Gender Gaps in Maths Become Skill Gaps?

Evidence from Ireland

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Abstract

How a gender gap in Maths is composed has different implications from how large it is. A gap driven mainly by measured skills points to different interpretations than a gap driven by differences in how similar skills translate into outcomes. This chapter studies that question using a within-cohort, same-outcome, two-measurement-point decomposition design. Using the Growing Up in Ireland longitudinal study, I decompose the gender gap in adjusted Leaving Certificate Maths points with predictors measured in late childhood (age 9) and early adolescence (age 13). The gap shifts from coefficient-dominant in childhood to endowment-dominant in adolescence: in the paired-sample timing test, the endowment share rises from 32% to 75%. An English comparison then serves as a subject-specific falsification test. There, the pattern reverses, with girls' advantage remaining coefficient-dominant even when boys hold stronger measured endowments. Taken together, the results suggest that the developmental stage at which the Maths gap is measured changes what the gap appears to consist of, and therefore the kinds of intervention it points toward.

Keywords: Gender gap; Maths achievement; Oaxaca-Blinder decomposition; Father absence; Cognitive skills; Socioeconomic background

JEL Codes: I21, I24, J12, J16

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1 Introduction

Where a gender gap in Maths exists, its structure may shift across developmental stages, and how it shifts matters as much as how large it is at any one age. If boys and girls differ mainly in measured skills, the gap may reflect earlier differences in preparation or learning. If they have similar measured skills but those skills map differently into outcomes, the gap may instead reflect differences in classroom processes, assessment, or subject-specific expectations. Existing work shows that these gaps vary across settings and developmental stages, that socioeconomic disadvantage reshapes them, and that boys may be particularly vulnerable to family instability in behavioural and educational outcomes (Autor et al., 2019; Bertrand & Pan, 2013; Borra et al., 2023; OECD, 2020). What is less well established is whether the decomposition structure of the same later outcome shifts when predictors are measured at different developmental stages.

This chapter studies that question using a within-cohort, same-outcome, two-measurement-point decomposition design. Using the Growing Up in Ireland (GUI) longitudinal study, I decompose the gender gap in Leaving Certificate Maths points using predictors measured at two points: late childhood (age 9, primary school) and early adolescence (age 13, secondary school). The contribution here is to compare the decomposition structure of the same later gap when predictors are measured at age 9 versus age 13. To do so, I use Oaxaca-Blinder decompositions, which distinguish between differences in observed characteristics and differences in the returns to those characteristics.

The decompositions reveal a compositional shift. In late childhood, the Maths gap is predominantly coefficient-dominant: boys and girls have similar measured characteristics, but those characteristics predict different outcomes. By early adolescence, the gap is mainly endowment-dominant. Boys have pulled ahead in measured numeracy, and that skill difference accounts for most of the gap. The chapter then subjects this result to a same-design comparison in English. There, the pattern reverses: girls retain a coefficient-dominant advantage even when boys hold stronger measured endowments. Read together, the Maths and English decompositions point to a subject-specific structure that reverses across the two outcomes.

I then extend the analysis to father absence. This is a secondary extension because the father-absence proxy is noisier and the relevant subsamples are smaller and more selected than in the main gender analysis. I document sizeable associated penalties tied to father absence. The longitudinal design, which observes parental presence repeatedly across waves, supports this descriptive extension while keeping interpretation appropriately cautious given the tighter measurement and attrition constraints in that subsample.

Ireland is a useful setting precisely because it is a relatively conservative case in which to study compositional change over time. Gender gaps in adolescent Maths are often modest in comparative perspective (Hyde & Mertz, 2009; Lindberg et al., 2010), so if the structure of the gap still shifts across developmental stages here, that possibility is unlikely to be unique

to high-gap contexts. The Irish system also offers consistent achievement measurement across subjects and cohorts through a centralised curriculum and standardised national examinations. The Growing Up in Ireland (GUI) study provides the longitudinal structure needed to compare the same later outcome using predictors measured at two earlier developmental stages.

Throughout, I treat the coefficients component as descriptive residual variation. I interpret it as capturing more than direct discrimination, and I read every result in the chapter as associational.

The main text presents the timing shift in Maths, supported by Figure 1 and distributional analyses, and uses the English decomposition as a subject-specific check. The father-absence extension is reported more cautiously. Detailed decomposition tables, OLS baselines, and robustness checks are provided in the appendices.

The rest of the chapter proceeds as follows. Section 2 reviews the related literature. Section 3 describes the data, measurement instruments, and sample construction. Section 4 presents the empirical strategy and results. Section 5 discusses the findings and their limits.

2 Related Literature

Three bodies of research motivate this study.

The first concerns the structure of gender gaps in Maths and the mechanisms that may underlie them. Cross-national evidence establishes that these gaps vary by institutional and cultural context (Fryer & Levitt, 2010; Guiso et al., 2008). Nollenberger et al. (2016) link variation in the Maths gap to gender-equality norms in the families' country of origin, showing that cultural environment shapes how the gap is expressed across settings. Within countries, Bertrand and Pan (2013) show that boys tend to outperform girls in Maths despite showing more behavioural problems and lower school engagement, while Autor et al. (2019) show that behavioural and schooling gaps widen with disadvantage. These patterns matter for more than the size of the gap alone. They also speak to whether observed characteristics differ between boys and girls, or whether similar characteristics are translated into outcomes differently across subjects and settings.

Other work points to mechanisms that can affect the coefficients side of a decomposition. Lavy and Sand (2018) show that teacher gender biases influence both performance and subject choices, an adult-side mechanism that maps directly onto how observed inputs translate into outcomes for boys and girls. Carlana (2019) document a student-side counterpart: implicit gender stereotypes among teachers widen the Maths gap and lower girls' confidence in their own ability. These mechanisms operate through the returns to characteristics rather than through differences in the characteristics themselves. The open question for this chapter is whether the same later outcome gap has a different decomposition structure when predictors are measured at different stages of development.

Skill formation and cumulative advantage motivate the second body of work. Cunha and Heckman (2007) formalise two mechanisms: self-productivity, where early skills reinforce later ones, and dynamic complementarity, where early investments raise the returns to subsequent investments. These mechanisms provide a conceptual framework for thinking about why coefficient differences at one stage might later appear as endowment differences. They are not tested directly here as causal mechanisms. They motivate the possibility that a gap which appears coefficient-dominant in childhood could become endowment-dominant by adolescence if earlier differences in how skills are rewarded accumulate over time. Empirical work shows that numeracy and verbal reasoning predict later achievement even after accounting for behavioural traits (Duncan et al., 2007), and that noncognitive characteristics such as self-regulation contribute independently to predictive power (Duckworth & Seligman, 2005).¹ The question for this

¹The size of that contribution has been contested. Duckworth and Seligman (2005) reported that self-discipline accounted for more than twice the variance in academic outcomes that IQ did, but direct replication of that specific finding is limited and the available replications are mixed. [vazsonyi2022does](#) find that intelligence is a significantly stronger predictor than self-control of teacher-reported academic competence and school grades over a two-year follow-up, and that only intelligence predicts developmental changes in academic performance. [michaelides2021self](#) report a preregistered replication finding that

chapter is how the decomposition roles of those inputs shift across developmental stages for a subject-specific outcome such as Maths.

The third strand concerns family structure and father absence. Father absence is associated with lower educational attainment and more behavioural difficulties, even conditional on background characteristics (McLanahan et al., 2013). Boys may be more vulnerable to household instability (Fomby & Cherlin, 2007; Lee & McLanahan, 2015), while girls' education may suffer more from reduced parental inputs and monitoring (Brenøe & Lundberg, 2018). More generally, children from disadvantaged backgrounds tend to fall further behind as they age (Bradley & Corwyn, 2002; Caro et al., 2009; Sirin, 2005). This literature speaks to a different question from the gender-timing question of the first two strands, but the same decomposition logic applies. The relevant question is whether a family-structure penalty in the same educational outcome is concentrated in measured characteristics or in their returns.

This chapter sits at the intersection of these three literatures. By decomposing the same outcome at two developmental stages, it addresses the timing question left open by the first strand and asks whether the observed compositional shift is descriptively consistent with the cumulative-advantage framework in the second. The cumulative-advantage framework is used here as conceptual scaffolding for interpreting the shift, not as a hypothesis tested by the decomposition itself. The English comparison provides a subject-specific check on whether the shift is peculiar to Maths, with method-related explanations as the main alternative the comparison helps adjudicate. The father-absence extension asks whether family structure compounds the main patterns under tighter measurement and sample constraints.

self-regulation, but not self-discipline, predicts academic achievement. **crede2017much**<empty citation> report a meta-analysis of the broader grit framework suggesting its predictive validity has been overstated. The Duckworth citation is used here for the more general claim that noncognitive characteristics contribute independently to academic outcomes, which the broader literature continues to support.

3 Data and Measurement

3.1 The Growing Up in Ireland Longitudinal Study

I use data from the Growing Up in Ireland study, the first large nationally representative longitudinal survey of children in Ireland. The Cohort '98 sample comprises 8,568 children born between November 1997 and October 1998, interviewed at ages 9 (Wave 1, 2007–08), 13 (Wave 2, 2011–12), 17/18 (Wave 3, 2015–16), and 20 (Wave 4, 2018–19). Retention was 87.8% at Wave 2, 72.6% at Wave 3, and 60.6% at Wave 4.²

The outcome is Leaving Certificate Maths performance at ages 17–18, reported retrospectively at Wave 4. Linked administrative Leaving Certificate records are not available to researchers in Ireland, so the analysis uses self-reported results from Wave 4 of GUI. Table 1 summarises the timing of data collection and the variables available at each wave.

Event	Date	Age (in years)	Variables of interest
Study-child is born	Nov/97 - Oct/98	0	-
Wave 1 data collection	Aug/07 - May/08	9	Reading and Maths logit scores, 4 SDQ scales, parental education (mother's and father's), income quintiles, co-educational school
Wave 2 data collection	Aug/11 - Mar/12	13	Verbal and numerical logit scores, BAS matrix reasoning, 4 SDQ scales, parental education (mother's and father's), income quintiles, school indicators (DEIS, co-educational, fee-paying, religious ethos)
Study-child sits the Junior Cert	Jun/13 - Jun/15	15–16	-
Wave 3 data collection	Apr/15 - Aug/16	17/18	Timing only; outcome not yet broadly observed
Study-child sits the Leaving Certificate	Jun/16 - Jun/17	17/18	-
Wave 4 data collection	Aug/18 - Jun/19	20	Leaving Certificate Maths points

Table 1: Timeline of events for the Growing Up in Ireland '98 Cohort (birth years 1997/98; baseline sample $n = 8,568$), showing survey waves, participant ages, and the main variables used in the analysis.

Outcome construction. The outcome is Leaving Certificate Maths points, self-reported at Wave 4 (age 20), two to three years after the examination.³ A 2017 reform changed the grading bands, so a dummy identifies which system applied to each participant.⁴ Where reported scores match known bonus-inclusive values (102, 105, 110, 113, 115, 120, or 125 points), 25 bonus points are subtracted; scores above 100 that cannot correspond to a valid bonus-inclusive Higher

²At Wave 4, 20-year-olds became the main respondents. This wave collected retrospective information on education, work, and time use, including self-reported Leaving Certificate results.

³At Wave 3, only 713 participants had already sat the Leaving Cert; most were still in school, so Wave 4 is the first wave that provides near-complete outcome coverage.

⁴Before 2017, bonus points were awarded for Higher Level Maths grades above 40%; the 2017 reform introduced broader H1–H8 / O1–O8 bands with a more uniform points structure.

Level grade are set to missing as part of the same correction step. The combined adjustment affects 894 observations (19.3% of non-missing raw Maths scores), and no observations above 100 remain after the correction. The resulting adjusted score is bounded between 0 and 100 and is used throughout, with 285 observations (6.2%) at the adjusted maximum of 100.

Outcome size and self-report concerns. With a pooled standard deviation of approximately 27 points, the raw gender gap of 4.4–5.2 points across the main decomposition samples corresponds to roughly 0.16–0.19 standard deviations, a small effect by conventional benchmarks (Hyde & Mertz, 2009). A further concern is differential reporting by gender: if boys systematically overreport and girls underreport performance, the observed gap would be mechanically widened. This possibility cannot be ruled out with survey data alone, but two features help to bound it. First, Appendix 21 shows that the reported score distribution follows the discrete grade-boundary structure of the Leaving Certificate instead of an arbitrary self-report pattern. Second, the English decomposition reverses sign relative to Maths, with girls outperforming boys, which a common-direction reporting bias that mechanically favours boys across subjects cannot easily explain. I treat asymmetric reporting as a standing measurement limitation, with caveats discussed alongside the main results.

Cognitive predictors. Cognitive skills at each wave are measured by standardised logit scores from age-appropriate assessments: reading and maths at Wave 1, and verbal reasoning, numerical ability, and BAS matrix reasoning at Wave 2. BAS matrix reasoning is a non-verbal cognitive measure.

Socioemotional predictors. Socioemotional traits are measured using the four difficulty subscales of the Strengths and Difficulties Questionnaire (SDQ): Emotional Symptoms, Conduct Problems, Hyperactivity/Inattention, and Peer Problems.⁵ The SDQ subscales are reported by the primary caregiver at both waves.

Family background. Family background is captured using parental education and equivalised household income. Parental education enters as dummies for Higher Secondary/Technical and Degree or Postgraduate, separately for each parent, with at most lower secondary education as the omitted category. No imputation is performed when father’s education is missing: those observations remain in the “no father’s education” specifications through a missing-father-education indicator and are dropped from the “with father’s education” specifications. Household income is equivalised using the Irish national (ESRI) scale (weights 1 / 0.66 / 0.33) and entered in quintiles.

⁵The Strengths and Difficulties Questionnaire is a widely used behavioural screening instrument for children and adolescents.

School context. School context is captured using indicators for Delivering Equality of Opportunity in Schools (DEIS) status, fee-paying status, religious ethos, and co-educational status. These are available at Wave 2, while only co-educational status is observed at Wave 1.

Variables available but not used in main specifications. Two further variables are available but kept out of the headline decompositions. Teacher reports of SDQ subscales at Wave 1 are used only in supplementary robustness analysis. Junior Certificate grades mapped to a 12-point scale are available but not used in the main specifications because both predictor and outcome are self-reported, creating a risk of correlated measurement error.

3.2 Sample construction and attrition

Table 2 summarises the sample flow from the original GUI cohort to the analytical samples used in each decomposition.

Table 2: Sample flow from GUI Cohort '98 to analytical samples

Stage	<i>N</i>	Attrition / restriction
<i>Panel A: Cohort follow-up</i>		
Wave 1 (age 9)	8,568	Original cohort
Wave 2 (age 13)	7,525	Non-response
Wave 3 (age 17/18)	6,216	Non-response
Wave 4 (age 20)	5,190	Non-response
Valid Leaving Certificate Maths score	4,333	Item-level missingness
<i>Panel B: Decomposition samples</i>		
Gender, W1 (no father's educ.)	3,690	Covariate missingness
Gender, W1 (with father's educ.)	3,241	+ father's education required
Gender, W2 (no father's educ.)	3,401	Covariate missingness
Gender, W2 (with father's educ.)	2,777	+ father's education required

Notes: Panel A reports sample retention across survey waves and outcome availability. Panel B reports the effective sample sizes used in the decomposition specifications.

Selection and attrition. Attrition is non-trivial and positively selected. Appendix 20 compares retained and lost cases using Wave 1 covariates. Retained children score about 0.3–0.4 standard deviations higher on baseline cognitive tests, display fewer externalising difficulties, and come from more educated, higher-income households, with similar imbalance patterns for boys and girls. The main gender decompositions speak most directly to the retained sample.

Inference and clustering. Standard errors are computed at the individual level. The GUI sample was drawn through schools, producing natural clustering at the school level, but school identifiers are available only through restricted-access Research Microdata Files. School-clustered standard errors are therefore replaced by a design-effect sensitivity analysis reported

in Appendix 16.

4 Empirical strategy and results

I organise the empirical analysis in three steps. Baseline OLS models establish the predictive structure. Oaxaca-Blinder decompositions then separate mean differences into endowments and coefficients. DFL reweighting shows where in the outcome distribution those differences are concentrated. Together, these steps move from association to composition to distribution.

4.1 Design

I estimate separate decompositions using predictors measured at age 9 (Wave 1) and age 13 (Wave 2), while holding the outcome fixed as Leaving Certificate Maths points at ages 17–18. The aim is to show whether the composition of the same gap looks different when observed at different stages of development.

To anchor the decomposition results, I first estimate OLS regressions of Leaving Certificate Maths points on cognitive, socioemotional, socioeconomic, and school predictors, separately for Wave 1 and Wave 2 (full tables in Appendix 7). Predictors measured closer to the outcome are more informative: Wave 2 models explain more variance (adjusted $R^2 \approx 0.39$ – 0.40) than Wave 1 models ($R^2 \approx 0.30$). Across waves, numerical ability, reading ability, hyperactivity, parental education, and father’s education missingness remain the most consistent correlates of performance.

Because cognitive tests change between waves and several school controls are only available at Wave 2, the same timing comparison is also re-estimated on a harmonised covariate set in the robustness analysis below and in Appendix 19.

For each wave, I estimate two models: one including a dummy for missing father’s education (using the full sample to capture paternal disengagement), and one controlling explicitly for father’s education level (using the subset with complete data).

4.2 Oaxaca-Blinder decomposition

To decompose the gender gap, I use the threefold Oaxaca-Blinder method (Blinder, 1973; Oaxaca, 1973):

$$\begin{aligned} \bar{Y}_G - \bar{Y}_B = & \underbrace{(\bar{\mathbf{X}}_G - \bar{\mathbf{X}}_B) \cdot \boldsymbol{\beta}_B}_{\text{Endowments}} + \underbrace{\bar{\mathbf{X}}_B \cdot (\boldsymbol{\beta}_G - \boldsymbol{\beta}_B)}_{\text{Coefficients}} \\ & + \underbrace{(\bar{\mathbf{X}}_G - \bar{\mathbf{X}}_B) \cdot (\boldsymbol{\beta}_G - \boldsymbol{\beta}_B)}_{\text{Interaction}} \end{aligned} \quad (4.1)$$

The endowments component measures how much of the gap is attributable to differences in observed characteristics between boys and girls, valued at boys' returns. The coefficients component measures how much is attributable to differences in returns to the same characteristics. The interaction captures their joint effect. I focus on the endowments and coefficients components in the main discussion, and report the interaction term in full in the figures and appendix tables.

Two interpretive cautions are necessary. First, the coefficients component is a residual. It captures genuine differences in returns alongside omitted variables, measurement error, and functional-form misspecification (Fortin et al., 2011; Jann, 2008). I interpret it descriptively, not as a measure of discrimination. Second, component magnitudes can be sensitive to the reference group (the “index number problem”). I use boys' coefficients as the reference and confirm robustness using the Neumark (1988) pooled-coefficient decomposition (Tables 8 and 9).

4.3 Gender gap decomposition

Figure 1 presents the main result: the gender gap in Maths decomposes differently depending on the age at which predictors are measured. At age 9, the gap is coefficient-dominant. In the full sample, boys outperform girls by 5.2 points, of which 1.57 points are attributable to endowments and 4.22 points to coefficients. In the sample restricted to observed father's education, the pattern is almost identical: a 4.4-point gap, with 1.30 points attributable to endowments and 3.64 points to coefficients.

By age 13, the same later outcome gap becomes mainly endowment-dominant. The total gap remains similar in size, 4.9 points in the full sample and 4.6 points in the observed-father-education sample, but almost all of it is now accounted for by endowments: 4.15 and 4.04 points respectively. The coefficients component falls to about 1.1–1.2 points and is no longer precisely estimated. In the paired-sample timing test reported later, the endowment share rises by 43–47 percentage points, from 32% at Wave 1 to 75% at Wave 2, which is statistically distinguishable from sampling variation.

Before unpacking what drives the Wave 2 endowment component, one caveat. The Wave 2 numerical ability measure is closer to the later Maths outcome than the Wave 1 numeracy measure, so this should be read as evidence about when the measured numeracy gap becomes visible in the developmental sequence. It does not isolate a separate causal mechanism.

The variable-level appendix tables show that the Wave 2 endowment component is driven mainly by measured cognitive differences, especially numerical ability and, to a lesser extent, verbal reasoning. In the full-sample Wave 2 decomposition, numerical ability alone contributes –3.66 points to the endowments component and verbal reasoning contributes a further –1.03 points, while boys' worse hyperactivity scores offset part of that advantage. The interpretation is that the same later gap becomes increasingly tied to measured skill differences, especially numeracy. Differences in how similar characteristics translate into outcomes play a smaller role by that stage.

Detailed variable-level decomposition tables are in Appendix 9 (Tables 22–23).

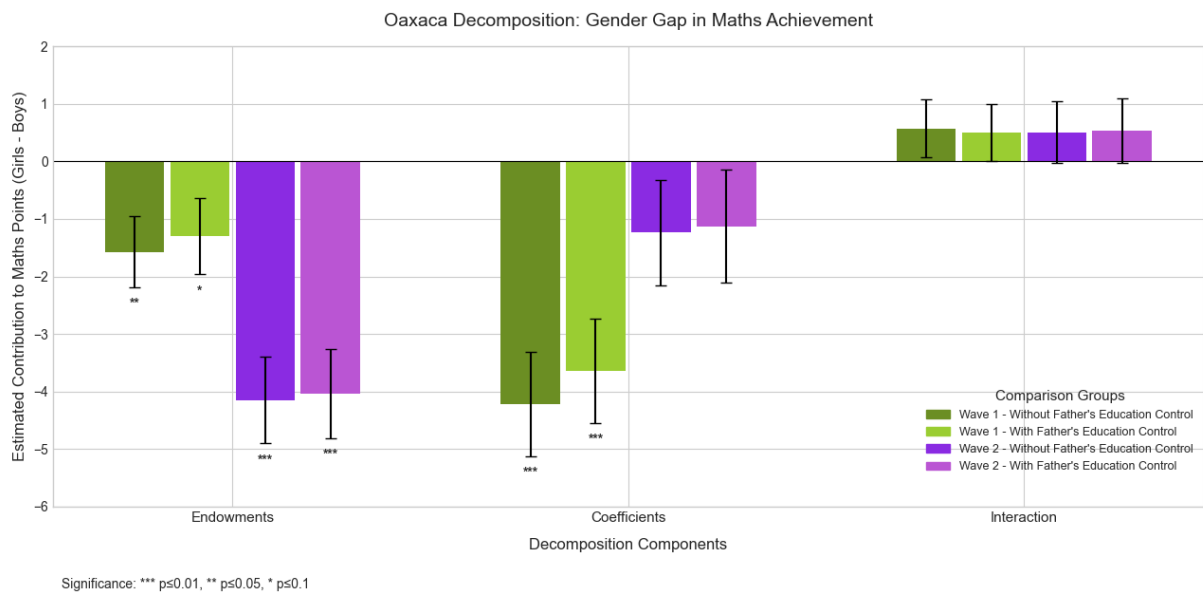


Figure 1: Oaxaca-Blinder decomposition of the gender gap in Leaving Certificate Maths by predictor wave and father-education specification. At age 9, the coefficient component is larger; at age 13, the endowment component is larger. Bars show endowment, coefficient, and interaction terms with bootstrap 95% confidence intervals (1,000 replications). Outcome: adjusted Leaving Certificate Maths points (0–100). Negative values are components associated with boys’ higher mean scores. Sample sizes are $n = 3,690, 3,241, 3,401,$ and $2,777$ across panels.

Figure 2 shows the same components as point estimates with confidence intervals, making the corresponding uncertainty easier to inspect.

4.4 A comparative test: the gender gap in English

If the coefficients component were primarily noise from omitted variables or functional-form misspecification, it should not flip sign systematically across subjects. To test this, I replicate the decomposition for Leaving Certificate English (Appendix 11).

The English results point in the opposite direction. Girls outperform boys by about 3–3.5 points, and the gap is mostly coefficient-dominant. At age 9, the coefficients component carries most of the gap; girls’ endowment difference is statistically indistinguishable from zero. By age 13, the pattern becomes sharper: boys hold a significant endowment advantage (–1.588 points, $p < 0.01$), yet girls continue to score higher because the coefficients component (+4.548 points) more than offsets that disadvantage. The same pattern holds under the Neumark pooled-coefficient benchmark: at Wave 2, endowments account for –40% of the English gap while coefficients account for 140% (Table 9), whereas the comparable Maths decomposition shows endowments at 80% (Table 8).

This cross-subject reversal is consistent with the coefficients component reflecting systematic,

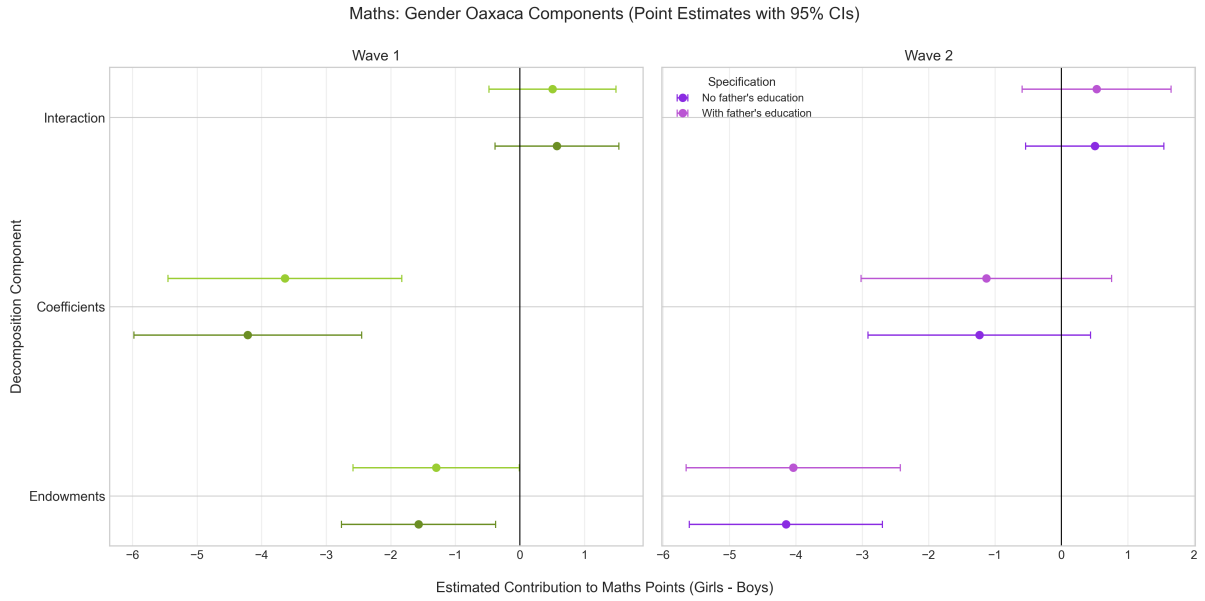


Figure 2: Coefficient-plot representation of the Oaxaca-Blinder decomposition of the gender gap in Leaving Certificate Maths. Points denote component estimates (endowments, coefficients, interaction) and horizontal whiskers denote 95% confidence intervals based on bootstrap standard errors (1,000 replications). Panels separate Wave 1 (age 9 predictors) and Wave 2 (age 13 predictors), and each panel reports estimates with and without father's education controls. Values to the left of zero indicate components associated with higher male scores. Estimates are descriptive and should not be interpreted as causal effects.

context-dependent differences in how observed characteristics map into outcomes. The contrast supports reading the coefficients component as a substantive feature of the data rather than a generic residual of the method.

4.5 Father absence

The next question is whether family structure compounds the patterns seen in the gender results. I treat this as a secondary extension because the father-absence proxy is noisier and the relevant subsamples are both smaller and more selected than the main gender sample.

Father absence is defined as consistent non-response to the father's questionnaire at both Wave 1 and Wave 2, intended to capture sustained paternal non-participation throughout childhood and early adolescence. This proxy combines structural absence and low engagement among resident partners. Wave 3 validation shows that the father-absent group is predominantly structurally absent (61.2% with no resident partner), with smaller shares of resident-but-disengaged (12.3%) and engaged-resident (6.0%) cases (Appendix 8, Table 21). Alternative definitions separating structural absence from resident disengagement preserve the baseline conclusion (Appendix 19, Table 40).

Father absence is strongly patterned and far from randomly distributed. Relative to father-present students, father-absent students have lower parental education, lower household income,

weaker cognitive scores, and more socioemotional difficulties (Table 14). They are also much more likely to attrit before Wave 4, with valid Leaving Certificate outcomes observed for 58% of father-absent students versus 73% of father-present students within the at-risk regression panel. The observed penalties should be interpreted against a background of substantial selection.

Even with that caution, the decomposition results show sizeable associated penalties for both boys and girls. In Wave 1, father absence is associated with a 13.6-point Maths penalty for boys and a 15.2-point penalty for girls. On the Neumark pooled-coefficient decomposition, endowments account for about 7.1 points of the boys' penalty and 7.7 points of the girls' penalty, while the coefficients component is about 6.4 points for boys and 7.6 points for girls. The corresponding endowment share is about 53% for boys and 50% for girls at Wave 1.

By Wave 2, the father-absence penalty remains large and the endowments component becomes more prominent for both genders. On the pooled-coefficient decomposition, endowments account for about 8.8 points of the boys' penalty and 8.9 points of the girls' penalty, while the coefficients component is about 4.8 points for boys and 6.3 points for girls. The corresponding endowment share rises to about 65% for boys and 58% for girls. The main takeaway is a sizeable descriptive penalty that operates through both measured characteristics and their returns, with coefficient-side contributions more visible for girls.

Figure 4 presents the same results visually by gender and wave, and Appendix 10 reports the variable-level decomposition tables.

4.6 Distributional decomposition

The Oaxaca-Blinder decompositions above focus on mean differences. To examine how gaps vary across the achievement distribution, I implement DiNardo-Fortin-Lemieux (DFL) reweighting decompositions (DiNardo et al., 1996) at the 10th, 25th, 50th, 75th, and 90th percentiles, with bootstrap confidence intervals from 1,000 stratified replications. I use DFL rather than the recentered influence function (RIF) method of Firpo et al. (2009) because Leaving Certificate Maths points exhibit mass points from the discrete grade-band structure, making the kernel density estimation required by RIF less reliable. Details are in Appendix 13.

Gender gap in Maths. Figure 3 and Table 3 show that the mean gap of 4–5 points hides a much larger median-region gap of 10–21 points, with very small gaps (0–5 points) at the 10th and 90th percentiles and often imprecise estimates in the tails. With age 9 predictors, the composition effect is close to zero across most quantiles. With age 13 predictors, it becomes materially larger, accounting for essentially all of the gap at the 25th percentile and 29% at the median. This is consistent with the mean-decomposition result that endowments account for a larger share of the gap by age 13.

Level selection in the Irish Maths examination. The distributional evidence has to be interpreted alongside level selection. In the Irish context, bonus points for Higher Level Maths have altered participation incentives and the profile of students taking the subject at that level (O’Meara et al., 2020). Support-split and extensive-intensive results indicate that part of the median contrast reflects differential placement between Higher and Ordinary Level Maths, not only within-level performance (Appendix 17; Appendix 19, Tables 38 and 39). The extensive-intensive decomposition separates whether the level-mix effect operates through the probability of sitting Higher Level Maths (extensive margin) or through points conditional on level (intensive margin). At Wave 2, endowments account for about 89% of the extensive-margin shift but only about 18–20% of the intensive-margin gap conditional on Higher Level participation. The timing result is not reducible to gender differences in level selection alone.

Table 3: DFL quantile decomposition of the gender gap in adjusted Leaving Certificate Maths points

Specification	Component	$q_{0.10}$	$q_{0.25}$	$q_{0.50}$	$q_{0.75}$	$q_{0.90}$
<i>Wave 1 predictors (age 9)</i>						
No father ed.	Total gap	0.0 (2.5)	5.0 (2.6)	10.0 (4.2)	7.0 (1.3)	4.0 (0.3)
	Composition	0.0 (0.4)	0.0 (0.8)	0.0 (0.5)	0.0 (1.9)	0.0 (0.3)
	Composition (%)	—	0%	0%	0%	0%
With father ed.	Total gap	5.0 (2.7)	5.0 (2.5)	14.0 (5.5)	7.0 (2.1)	4.0 (0.4)
	Composition	0.0 (2.4)	0.0 (1.8)	0.0 (2.0)	4.0 (2.0)	0.0 (0.6)
	Composition (%)	0%	0%	0%	57%	0%
<i>Wave 2 predictors (age 13)</i>						
No father ed.	Total gap	5.0 (1.8)	5.0 (2.0)	14.0 (5.2)	7.0 (1.4)	4.0 (0.4)
	Composition	5.0 (2.5)	5.0 (1.6)	4.0 (3.0)	4.0 (1.5)	0.0 (1.2)
	Composition (%)	100%	100%	29%	57%	0%
With father ed.	Total gap	0.0 (2.2)	2.0 (2.3)	21.0 (5.4)	3.0 (1.7)	4.0 (1.0)
	Composition	0.0 (2.1)	2.0 (2.2)	6.0 (4.5)	0.0 (1.7)	0.0 (1.8)
	Composition (%)	—	100%	29%	0%	0%

Notes: Standard errors in parentheses are based on 1,000 stratified bootstrap replications. At each quantile, *Total gap* is boys minus girls. *Composition* is the counterfactual-minus-girls component. The residual *Structure* component is not shown, so that $Total = Composition + Structure$. *Composition (%)* is defined as $Composition/Total$ and is omitted when the total gap is zero or close to zero. “No father ed.” includes a missing-father-education indicator. “With father ed.” restricts the sample to observations with observed father’s education. The outcome is adjusted Leaving Certificate Maths points (0–100). Sample sizes vary across specifications.

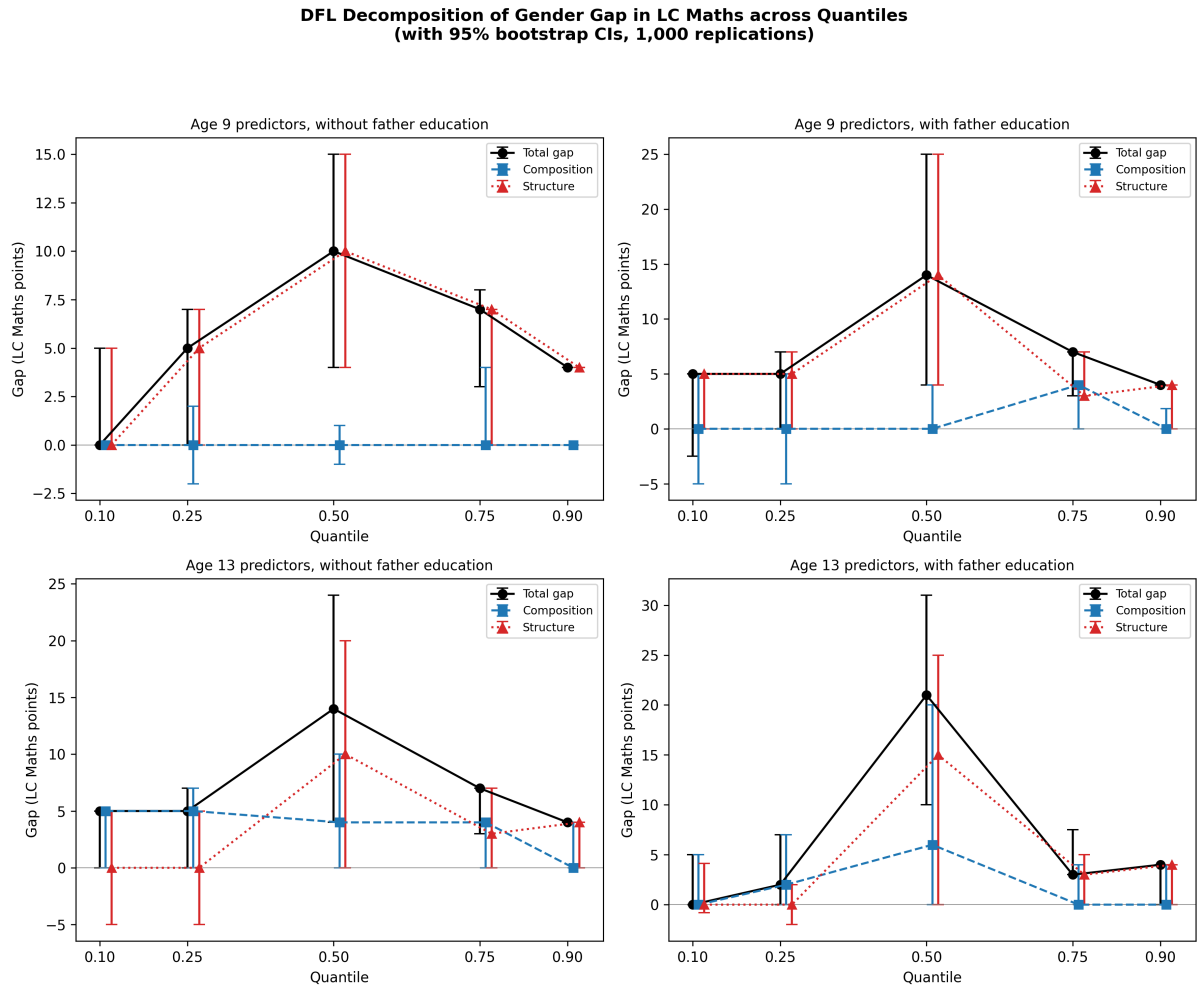


Figure 3: DFL decomposition of the gender gap in Leaving Certificate Maths across quantiles, with 95% bootstrap confidence intervals (1,000 stratified replications). Top row: age 9 predictors; bottom row: age 13 predictors; left column excludes father's education; right column includes it. Black circles: total gap (boys – girls). Blue squares: composition effect (counterfactual – girls). Red triangles: structure effect (boys – counterfactual). The gender gap is concentrated at the median across all specifications, with composition effects becoming more important when age 13 predictors are used. Sample: $N = 4,333$ (complete cases vary by specification). Outcome: adjusted Leaving Certificate Maths points (0–100 scale).

Gender gap in English. The DFL decomposition for English (Table 4) mirrors the Oaxaca-Blinder reversal. The mean gap (3.1 points favouring girls) is concentrated at the 10th percentile (10 points with age 9 predictors) and is negligible from the median upward. The English gap therefore lives in the lower tail, while the Maths gap lives in the middle of the distribution. With age 13 predictors, the composition effect turns negative at the mean (–1.49 points, $SE = 0.50$): boys have better measured characteristics on average, yet girls still score higher. The coefficient component accounts for 151% of the mean gap at age 13, consistent with the subject-specific interpretation from the Oaxaca-Blinder analysis. Detailed figures and tables are in Appendix 14.

Table 4: DFL quantile decomposition of the gender gap in adjusted Leaving Certificate English points

Specification	Component	$q_{0.10}$	$q_{0.25}$	$q_{0.50}$	$q_{0.75}$	$q_{0.90}$
<i>Wave 1 predictors (age 9)</i>						
No father ed.	Total gap	10.0 (3.6)	4.0 (1.0)	0.0 (1.7)	0.0 (1.1)	0.8 (1.0)
	Composition	2.0 (2.6)	0.0 (0.5)	0.0 (1.1)	0.0 (0.7)	0.0 (0.1)
	Composition (%)	20%	0%	—	—	0%
With father ed.	Total gap	9.0 (2.2)	4.0 (1.4)	5.0 (2.7)	0.0 (2.4)	2.0 (0.9)
	Composition	0.0 (1.9)	0.0 (0.8)	0.0 (1.0)	0.0 (0.5)	0.0 (0.2)
	Composition (%)	0%	0%	0%	—	0%
<i>Wave 2 predictors (age 13)</i>						
No father ed.	Total gap	8.0 (2.0)	4.0 (1.7)	0.0 (2.5)	0.0 (2.3)	2.0 (1.0)
	Composition	-2.5 (2.8)	0.0 (1.2)	-4.0 (2.0)	-3.0 (1.4)	0.0 (0.3)
	Composition (%)	-31%	0%	—	—	0%
With father ed.	Total gap	6.0 (1.9)	9.0 (2.8)	5.0 (1.0)	5.0 (2.0)	2.0 (0.9)
	Composition	-3.0 (1.8)	0.0 (1.8)	0.0 (1.7)	0.0 (1.4)	0.0 (0.4)
	Composition (%)	-50%	0%	0%	0%	0%

Notes: Standard errors in parentheses are based on 1,000 stratified bootstrap replications. At each quantile, *Total gap* is girls minus boys, so positive values indicate a girls' advantage. *Composition* is the counterfactual-minus-boys component. The residual *Structure* component is not shown, so that $Total = Composition + Structure$. *Composition (%)* is defined as $Composition/Total$ and is omitted when the total gap is zero or close to zero. Negative composition values indicate endowments that favour boys, holding the structure component aside. "No father ed." includes a missing-father-education indicator. "With father ed." restricts the sample to observations with observed father's education. The outcome is adjusted Leaving Certificate English points. Sample sizes vary across specifications.

Father-absence gaps in Maths. The father-absence penalty in Maths is broadly distributed across quantiles rather than concentrated at the median. Boys show the largest gap at the median (-28.5 points, $SE = 6.3$), with the composition effect explaining 88–91%. For girls, the gap peaks at the 75th percentile (-29 points), where the composition share is stable at about 28% across waves. At the median, the composition effect for girls varies sharply by wave: 11% with age 9 predictors and 61% with age 13 predictors.

Father-absence gaps in English. The father-absence penalty in English is smaller but shows a distinctive feature for girls. With age 13 predictors, the composition effect (-5.52 points) exceeds the total gap (-4.18 points). Measured-skill differences between father-present and father-absent girls more than account for the overall English penalty, with the residual structure component partially offsetting the observed gap. Bootstrap tables for both subjects are reported in Appendix 13 (Maths) and Appendix 15 (English).

The distributional evidence reinforces the mean-decomposition pattern: by age 13, measured skill differences carry most of the Maths gap, while the English gap remains coefficient-dominant across the distribution.

4.7 Robustness

Paired-bootstrap timing test. I assess the timing claim with a paired bootstrap that resamples the same individuals across waves and anchors the comparison in within-sample changes, with less dependence on across-sample composition. Table 5 shows that, in the primary Maths specification, the endowment component rises by $+2.51$ points from Wave 1 to Wave 2 (95% CI: $[1.52, 3.50]$, $p < 0.001$), and the endowment share increases by 43 percentage points, from 32% at Wave 1 to 75% at Wave 2. The same paired-bootstrap test applied to English shows the opposite shift: the endowment component falls by -1.58 points ($p < 0.001$). The cross-subject contrast holds as a formal timing test, not just as a descriptive comparison.

Father absence and the timing claim. For father-absence decompositions in Maths, the shifts are directionally similar but statistically imprecise ($p = 0.21$ – 0.34). The father-absence extension does not carry a timing claim. Its contribution is to document a sizeable mean penalty (14–15 points) that remains visible under attrition sensitivity checks, separate from the compositional shift across developmental stages that the gender results establish.

Quantile-level timing test. The quantile-level version of the same test, reported in Table 6, shows that the timing shift is visible across the distribution rather than only at the mean. In Maths, the no-father-education specification shows significant positive composition shifts at the 10th ($+5.00$, $p = 0.045$) and 25th percentiles ($+5.00$, $p = 0.005$), with the mean shift marginally significant ($+2.10$, $p = 0.052$). In English, the mean composition shift is -1.98

Table 5: Paired bootstrap timing test of Wave 1 and Wave 2 decomposition differences

Specification	Endowment Δ	95% CI	p	Share Δ (pp)	N
Maths, no father ed.	+2.51***	[1.52, 3.50]	0.000	+43.1***	2,964
Maths, with father ed.	+2.34***	[1.15, 3.52]	0.000	+46.5***	2,330
English, no father ed.	-1.58***	[-2.19, -0.96]	0.000	-53.8***	2,944
English, with father ed.	-1.34***	[-2.04, -0.66]	0.000	-36.7***	2,316
FA Maths, boys	+1.67	[-0.82, 4.08]	0.208	+12.3	1,314
FA Maths, girls	+1.21	[-1.35, 3.42]	0.336	+7.9	1,292
FA English, boys	+0.22	[-1.28, 1.84]	0.798	+3.9	1,300
FA English, girls	+1.28*	[-0.13, 2.56]	0.072	+30.7*	1,289

Notes: Paired bootstrap test of H_0 : $\text{component}_{w2} = \text{component}_{w1}$, using standard Oaxaca-Blinder decompositions with boys' coefficients as the reference. The same individuals are resampled in each replication, which preserves within-person dependence across waves. Results are based on 1,000 bootstrap replications with 95% percentile confidence intervals. Δ denotes the Wave 2 estimate minus the Wave 1 estimate. Share Δ reports the change in the endowment share of the total gap, in percentage points. The coefficient-component change is the exact complement, $-\Delta$, by construction.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

($p = 0.002$), with significant negative shifts also visible at the 50th and 75th percentiles. The quantile evidence reinforces the mean-based timing test rather than substituting for it.

Harmonised covariate specification. A natural concern is that the Wave 1 to Wave 2 shift reflects changing predictor sets instead of developmental timing. Cognitive tests differ across waves (BAS-derived measures at age 9; Drumcondra Numerical Ability and Verbal Reasoning at age 13), and several school controls are only available at Wave 2. To address this, I re-estimate the decomposition using only conceptually matched covariates that exist at both waves: a numeracy measure, a reading/verbal measure, the four SDQ subscales, primary caregiver education, equivalised household income, school type, and a father-education-missing indicator. With these harmonised covariate sets, the endowment-share shift remains large and statistically significant: approximately 51–62 percentage points in Maths, with a reversed shift of about 36–54 percentage points in English (Appendix 19, Tables 36 and 37). The timing pattern is not an artefact of richer Wave 2 measurement. It survives when the predictor structure is held constant across waves. One standing limitation remains: harmonised covariates address composition, but the age 13 cognitive tests may also be more informative measures of Leaving Certificate Maths than the age 9 tests independently of developmental timing. The design therefore remains descriptive and associational.

Reference-group sensitivity. The Neumark (1988) pooled-coefficient decomposition confirms that results are robust to the reference-group choice across all specifications (Tables 8–11). For the headline Maths comparison, the Neumark endowment share rises from 28% at Wave 1 to 80% at Wave 2, matching the direction and broad magnitude of the standard Oaxaca result. Table 7 consolidates the evidence from all robustness checks for the main gender-timing result.

Table 6: Quantile-level timing test of Wave 2 minus Wave 1 differences in DFL components

Quantile	Composition Δ			Structure Δ		
	Est.	SE	p	Est.	SE	p
<i>Gender, Maths (no father ed.)</i>						
Mean	+2.10*	1.08	0.052	-2.43*	1.41	0.084
0.10	+5.00**	2.50	0.045	+0.00	3.65	1.000
0.25	+5.00***	1.79	0.005	-5.00	3.21	0.120
0.50	+4.00	3.04	0.188	+0.00	6.78	1.000
0.75	+4.00	2.44	0.101	-4.00	2.46	0.104
0.90	+0.00	1.27	1.000	+0.00	1.35	1.000
<i>Gender, Maths (with father ed.)</i>						
Mean	+1.91	1.17	0.103	-1.71	1.51	0.258
0.10	+0.00	3.22	1.000	-5.00*	2.87	0.082
0.25	+2.00	2.85	0.482	-5.00*	2.61	0.055
0.50	+6.00	4.91	0.221	+1.00	8.39	0.905
0.75	-4.00	2.61	0.125	+0.00	1.84	1.000
0.90	+0.00	1.95	1.000	+0.00	2.11	1.000
<i>Gender, English (no father ed.)</i>						
Mean	-1.98***	0.65	0.002	+1.80*	0.92	0.051
0.10	-4.49	3.87	0.246	+2.49	4.92	0.613
0.25	+0.00	1.34	1.000	+0.00	2.18	1.000
0.50	-4.00*	2.26	0.077	+4.00	3.35	0.233
0.75	-3.00*	1.60	0.062	+3.00	2.87	0.295
0.90	+0.00	0.34	1.000	+1.20	1.40	0.391
<i>Gender, English (with father ed.)</i>						
Mean	-1.68**	0.66	0.011	+1.25	0.98	0.205
0.10	-3.00	2.66	0.259	+0.00	3.04	1.000
0.25	+0.00	2.00	1.000	+5.00*	2.75	0.069
0.50	+0.00	2.01	1.000	+0.00	3.14	1.000
0.75	+0.00	1.48	1.000	+5.00	3.29	0.128
0.90	+0.00	0.41	1.000	+0.00	1.21	1.000

Notes: Approximate z -tests use $SE_{\Delta} = \sqrt{SE_{W1}^2 + SE_{W2}^2}$, which is conservative under positive correlation. Δ denotes the Wave 2 estimate minus the Wave 1 estimate. Results are based on 1,000 bootstrap replications per wave. Positive composition Δ indicates that the composition component explains more of the gap at age 13 than at age 9. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 7: Robustness summary for the main gender-timing result in Leaving Certificate Maths

Check	Evidence for the Wave 1 $\rightarrow$ Wave 2 shift
Standard Oaxaca–Blinder (paired-sample timing test)	32% at Wave 1 versus 75% at Wave 2; paired-bootstrap shift of +43 percentage points ($p < 0.001$).
Neumark pooled-coefficient decomposition	28% at Wave 1 versus 80% at Wave 2; confirms that the timing pattern is not driven by the reference group.
Harmonised wave covariates	Matched covariate blocks still yield a large shift: about +51 to +62 percentage points in Maths, with a reversed shift of about 36–54 percentage points in English.
Level-mix decomposition (extensive vs intensive)	At Wave 2, endowments dominate the extensive margin (about 89%), while gaps conditional on Higher Level participation are smaller and less endowment-led (about 18–20%).
School-clustering inference sensitivity	With design-effect inflation ($\times 1.3$ and $\times 1.5$), Wave 1 remains coefficient-led and Wave 2 remains endowment-led.

Notes: Entries summarise Tables 5, 8, 36, 37, 38, 39, and 32. All figures are descriptive decomposition results.

Table 8: Neumark pooled-coefficient decomposition of adjusted Leaving Certificate Maths points

	Gap	Endowments (pooled $\hat{\beta}$)	Coefficients (unexplained)	Endowment (%)
Maths, W1 no father ed.	5.212*** (0.922)	1.469*** (0.534)	3.744*** (0.873)	28%
Maths, W1 with father ed.	4.434*** (1.029)	1.318** (0.625)	3.116*** (0.806)	30%
Maths, W2 no father ed.	4.882*** (1.022)	3.907*** (0.676)	0.975 (0.777)	80%
Maths, W2 with father ed.	4.635*** (1.204)	3.876*** (0.725)	0.759 (0.845)	84%

Notes: Neumark (1988) pooled-coefficient decomposition. The pooled OLS coefficient vector is used as the reference. Endowments are defined as $(\bar{X}_A - \bar{X}_B)\hat{\beta}_{\text{pool}}$. Coefficients denote the unexplained component. Bootstrap standard errors are shown in parentheses (100 replications). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 9: Neumark pooled-coefficient decomposition of adjusted Leaving Certificate English points

	Gap	Endowments (pooled $\hat{\beta}$)	Coefficients (unexplained)	Endowment (%)
English, W1 no father ed.	3.079*** (0.636)	0.666* (0.399)	2.413*** (0.518)	22%
English, W1 with father ed.	3.566*** (0.607)	0.776** (0.345)	2.790*** (0.552)	22%
English, W2 no father ed.	2.906*** (0.678)	-1.169*** (0.404)	4.075*** (0.503)	-40%
English, W2 with father ed.	3.135*** (0.733)	-0.815* (0.434)	3.950*** (0.613)	-26%

Notes: Neumark (1988) pooled-coefficient decomposition. The pooled OLS coefficient vector is used as the reference. Endowments are defined as $(\bar{X}_A - \bar{X}_B)\hat{\beta}_{\text{pool}}$. Coefficients denote the unexplained component. Bootstrap standard errors are shown in parentheses (100 replications). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 10: Neumark pooled-coefficient decomposition of adjusted Leaving Certificate Maths points by father absence

	Gap	Endowments (pooled $\hat{\beta}$)	Coefficients (unexplained)	Endowment (%)
Maths, W1, boys	13.564*** (2.847)	7.127*** (1.762)	6.438*** (2.429)	53%
Maths, W1, girls	15.225*** (2.687)	7.672*** (1.720)	7.553*** (2.123)	50%
Maths, W1, all	14.736*** (2.073)	7.709*** (1.281)	7.026*** (1.592)	52%
Maths, W2, boys	13.564*** (2.849)	8.793*** (1.848)	4.771** (2.245)	65%
Maths, W2, girls	15.225*** (2.440)	8.881*** (1.917)	6.344*** (1.701)	58%
Maths, W2, all	14.736*** (1.829)	9.215*** (1.326)	5.520*** (1.490)	63%

Notes: Neumark (1988) pooled-coefficient decomposition. The pooled OLS coefficient vector is used as the reference. Endowments are defined as $(\bar{X}_A - \bar{X}_B)\hat{\beta}_{\text{pool}}$. Coefficients denote the unexplained component. Bootstrap standard errors are shown in parentheses (100 replications). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 11: Neumark pooled-coefficient decomposition of adjusted Leaving Certificate English points by father absence

	Gap	Endowments (pooled $\hat{\beta}$)	Coefficients (unexplained)	Endowment (%)
English, W1, boys	5.673*** (2.025)	3.760*** (1.019)	1.913 (1.707)	66%
English, W1, girls	4.179** (1.660)	3.408*** (0.968)	0.770 (1.446)	82%
English, W1, all	4.621*** (1.352)	3.256*** (0.766)	1.365 (1.011)	70%
English, W2, boys	5.673*** (2.058)	3.979*** (1.087)	1.694 (1.568)	70%
English, W2, girls	4.179** (1.774)	4.692*** (0.915)	-0.514 (1.391)	112%
English, W2, all	4.621*** (1.326)	4.035*** (0.581)	0.586 (1.070)	87%

Notes: Neumark (1988) pooled-coefficient decomposition. The pooled OLS coefficient vector is used as the reference. Endowments are defined as $(\bar{X}_A - \bar{X}_B)\hat{\beta}_{\text{pool}}$. Coefficients denote the unexplained component. Bootstrap standard errors are shown in parentheses (100 replications). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Selection: IPW and Lee bounds. IPW reweighting finds no evidence of selection bias on observables: the unweighted and reweighted estimates are essentially identical at reported precision (median absolute change 0.0% across 12 specifications), with all 12 specifications preserving signs and magnitudes (Table 12). In practice, the trimmed and normalised observability weights are tight enough that the reweighted decompositions are almost identical to the unweighted ones at the reported precision. Lee bounds for the father-absence Maths gap remain negative in all samples ($[-30.55, -5.73]$ overall; Appendix 18). For the main gender decomposition, Lee bounding is less informative because attrition rates do not differ significantly by gender, so the implied bounds add little beyond the retained-sample analysis itself.

Selectivity caveat. The overall sample loss from the baseline cohort (57–68% depending on specification) means results represent children who remained in the study through age 20, who are systematically more advantaged. All findings are interpreted conditional on this selected sample.

Clustering and design effects. Standard errors are computed at the individual level. School identifiers are available only through restricted-access Research Microdata Files, so school-clustered standard errors are not implemented in the headline tables. A design-effect sensitivity analysis (Appendix 16, Table 32) inflates standard errors at design-effect factors of $\times 1.3$ and $\times 1.5$. Across this range, Wave 1 remains coefficient-led and Wave 2 remains endowment-led

Table 12: Inverse-probability-weighted Oaxaca–Blinder sensitivity to outcome observability

Specification	Endowments		Coefficients		Change (%)	
	UW	IPW	UW	IPW	Endow.	Coef.
G Maths, W1	1.04*	1.04	3.71***	3.71***	0.0	0.0
G Maths, W2	3.60***	3.60***	0.95	0.95	0.0	0.0
G English, W1	0.91**	0.91**	2.97***	2.97***	0.0	0.0
G English, W2	-1.45***	-1.45***	5.00***	5.00***	0.0	0.0
FA Maths, W1, boys	6.95***	6.95***	6.61***	6.61**	0.0	0.0
FA Maths, W1, girls	7.77***	7.77***	7.46***	7.46***	0.0	0.0
FA Maths, W2, boys	8.57***	8.57***	4.99*	4.99*	0.0	0.0
FA Maths, W2, girls	8.87***	8.87***	6.35***	6.35***	0.0	0.0
FA English, W1, boys	3.74***	3.74***	1.93	1.93	0.0	0.0
FA English, W1, girls	3.44***	3.44***	0.74	0.74	0.0	0.0
FA English, W2, boys	3.70***	3.70***	1.97	1.97	0.0	0.0
FA English, W2, girls	4.86***	4.86***	-0.68	-0.68	0.0	0.0
Sign stability	Endowments: 12/12; Coefficients: 12/12					
Median $ \Delta $	Endowments: 0.0%; Coefficients: 0.0%					

Notes: UW denotes the unweighted Oaxaca–Blinder decomposition and IPW the inverse-probability-weighted decomposition. The selection model is a logit for observability of a valid Leaving Certificate score using Wave 1 covariates. Weights are trimmed at the 1st and 99th percentiles and normalised. Bootstrap standard errors are based on 100 replications.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

for the main gender-timing result.

5 Discussion and conclusion

The main finding is that the composition of the gender gap in Maths differs between childhood and early adolescence. At age 9, boys and girls look similar on measured skills, but the same characteristics predict different outcomes. By age 13, boys hold a measured numeracy advantage, and that difference accounts for most of the gap. The gap persists, but its composition changes. The findings below organise what is central to interpreting this pattern.

Compositional shift. The decompositions show a clear transition from a coefficient-dominant gap at age 9 to an endowment-dominant gap at age 13, and this shift survives harmonised covariate specifications, alternative decomposition methods, and a paired bootstrap test. The English comparison then functions as a subject-specific check. In English, the pattern reverses entirely, with girls' advantage remaining coefficient-dominant even when boys hold stronger measured endowments. That reversal makes a purely residual interpretation of the coefficients component harder to sustain and instead points to subject-specific differences in how observed characteristics map into outcomes.

Father absence and gendered channels. Father absence is descriptively associated with large penalties, but through different decomposition channels depending on gender. The 14–15 point achievement gap operates through both endowments and coefficients, with the composition differing by gender. For boys, both lower measured skills and the coefficients component contribute. For girls, coefficient-side contributions are more visible, especially in the earlier-wave decomposition, with family-resource-related terms playing a more prominent role (Brenøe & Lundberg, 2018). Given proxy construction and attrition constraints, this extension is interpreted as supportive descriptive evidence and is kept secondary to the chapter's core result.

Distributional concentration. The mean gap masks distributional concentration at the median. The gender gap reaches 10–21 points around the median while remaining comparatively modest at both tails. Average-based summaries can misstate where the largest differences are; interventions focused only on the extremes would miss the centre of the distribution.

5.1 Mechanisms

Several overlapping channels are consistent with the observed patterns. The earliest of these, teacher expectations and implicit biases, may shape how boys' and girls' observed skills are rewarded long before later skill differences become visible (Carlana, 2019; Lavy & Sand, 2018). As children move into adolescence, stereotype threat and self-efficacy gaps may compound the early differential treatment by lowering girls' Maths confidence and engagement (Cimpian et al., 2016; Spencer et al., 1999; Whitcomb et al., 2020). These channels are likely interconnected.

One possibility is that early coefficient differences feed into later skill differences through a cumulative process of reinforcement and skill accumulation, consistent with the general logic of Cunha and Heckman (2007). The decomposition design cannot identify which of these channels is operating or establish the causal sequence directly.

5.2 Limitations

Three categories of limitation shape interpretation.

Identification. The design is associational. All findings should be interpreted as descriptive decompositions, not causal effects. The coefficients component is a residual that captures omitted variables, measurement differences, and model misspecification alongside genuine differences in returns. IPW reweighting on observables (Table 12) provides a partial check on selection, but residual selection on unobservables cannot be excluded.

Inference. School-clustered standard errors are not available because school identifiers require restricted access. In school-based surveys, within-school dependence can inflate standard errors relative to iid assumptions (Hedges & Hedberg, 2007; Juras, 2016). Design-effect sensitivity exercises confirm that the main conclusions survive plausible clustering adjustments (Appendix 16). GUI wave-specific survey weights exist, but consistent weights across all four decomposition samples (which span Waves 1, 2, and 4 with varying missingness patterns) are not available in the public-use file. Results are unweighted conditional associations.

Measurement: outcome and recall. The outcome is self-reported Leaving Certificate Maths scores collected two to three years after the exam, so recall error and social-desirability bias remain possible. A specific concern is asymmetric reporting by gender: if boys systematically overreport and girls underreport, the observed gap would be mechanically widened. A reporting-diagnostic analysis (Appendix 21) confirms that the Leaving Certificate Maths distribution reflects grade-boundary discretisation and does not look like free self-report, which limits the scope for idiosyncratic misreporting.

Measurement: level selection. The score amalgamates Higher and Ordinary Level results, and gendered level selection may inflate distributional contrasts at the median (Appendix 17).

Measurement: father-absence proxy. The father-absence proxy (non-response at both waves) mixes structural absence, resident disengagement, and missing data (Appendix 8, Table 21).

Selection and attrition. Retained children score 0.3–0.4 standard deviations higher on Wave 1 cognitive tests and come from more educated, higher-income households than those lost to

attrition. The pattern is similar for boys and girls (Appendix 20). Father-absent students are significantly more likely to attrit (58% valid outcomes vs. 73% for father-present, within the at-risk regression panel), so the analytical sample over-represents more advantaged father-absent students. IPW reweighting and Lee bounds both indicate that the father-absence penalty survives under selection on observables and monotonic attrition assumptions, but residual selection on unobservables cannot be ruled out.

Despite these limitations, the key patterns replicate across decomposition methods (standard O-B, Neumark, DFL), robustness checks (IPW, Lee bounds, harmonised wave comparisons, design-effect inflation), and outcome definitions. The coefficient-dominant gap at age 9, the endowment-dominant gap at age 13, the subject-specific reversal in English, and the gendered channels of father-absence effects all remain visible across these checks.

5.3 Implications

These findings have economic significance beyond the classroom. Maths achievement is a strong predictor of STEM entry, and the early gaps documented here may contribute to women's under-representation in fields where wage returns are high (Card & Payne, 2021). More broadly, the macroeconomic literature on gender and talent allocation suggests that early attainment gaps can have downstream consequences for later occupational sorting, even though the present chapter does not estimate those aggregate effects directly.

The timing result suggests directions for intervention design. If the gap is primarily coefficient-dominant at age 9, then programmes that address how boys' and girls' observed characteristics translate into outcomes may warrant investigation during middle childhood, before the decomposition becomes endowment-dominant. Interventions aimed at classroom expectations, feedback, and related noncognitive margins may be especially relevant at that stage (Alan & Ertac, 2018; Lavy & Sand, 2018). The distributional concentration at the median further suggests that middle-performing girls may be a policy-relevant group to study. These are hypotheses for causal evaluation, not conclusions the current design can establish.

Taken together, the chapter shows that the developmental stage at which the gender gap in Maths is measured changes what the gap appears to consist of, and that the English reversal makes a generic decomposition-method explanation hard to sustain. The findings generate a more specific set of hypotheses about when and where intervention might matter, but whether earlier intervention during primary school would narrow the gap remains a causal question this design cannot answer.

References

- Alan, S., & Ertac, S. (2018). Fostering Patience in the Classroom: Results from a Randomized Educational Intervention. *Journal of Political Economy*, 126(5), 1865–1911.
- Autor, D., Figlio, D., Karbownik, K., Roth, J., & Wasserman, M. (2019). Family Disadvantage and the Gender Gap in Behavioral and Educational Outcomes. *American Economic Journal: Applied Economics*, 11(3), 338–381.
- Bertrand, M., & Pan, J. (2013). The Trouble with Boys: Social Influences and the Gender Gap in Disruptive Behavior. *American Economic Journal: Applied Economics*, 5(1), 32–64.
- Blinder, A. S. (1973). Wage Discrimination: Reduced Form and Structural Estimates. *Journal of Human Resources*, 8(4), 436–455.
- Borra, C., Iacovou, M., & Sevilla, A. (2023). Adolescent Development and the Math Gender Gap. *European Economic Review*, 158, 104542. <https://doi.org/10.1016/j.euroecorev.2023.104542>
- Bradley, R. H., & Corwyn, R. F. (2002). Socioeconomic Status and Child Development. *Annual Review of Psychology*, 53(1), 371–399.
- Brenøe, A. A., & Lundberg, S. (2018). Gender Gaps in the Effects of Childhood Family Environment: Do They Persist into Adulthood? *European Economic Review*, 109, 42–62.
- Card, D., & Payne, A. A. (2021). High School Choices And The Gender Gap In STEM. *Economic Inquiry*, 59(1), 9–28.
- Carlana, M. (2019). Implicit Stereotypes: Evidence from Teachers' Gender Bias. *The Quarterly Journal of Economics*, 134(3), 1163–1224.
- Caro, D. H., McDonald, J. T., & Willms, J. D. (2009). Socio-Economic Status and Academic Achievement Trajectories from Childhood to Adolescence. *Canadian Journal of Education*, 32(3), 558–590.
- Cimpian, J. R., Lubienski, S. T., Timmer, J. D., Makowski, M. B., & Miller, E. K. (2016). Have Gender Gaps in Math Closed? Achievement, Teacher Perceptions, and Learning Behaviors Across Two ECLS-K Cohorts. *AERA Open*, 2(4), 1–19.
- Cunha, F., & Heckman, J. (2007). The Technology of Skill Formation. *American Economic Review*, 97(2), 31–47.
- DiNardo, J., Fortin, N. M., & Lemieux, T. (1996). Labor Market Institutions and the Distribution of Wages, 1973-1992: A Semiparametric Approach. *Econometrica*, 64(5), 1001–1044.
- Duckworth, A. L., & Seligman, M. E. (2005). Self-Discipline Outdoes IQ in Predicting Academic Performance of Adolescents. *Psychological Science*, 16(12), 939–944.
- Duncan, G. J., Dowsett, C. J., Claessens, A., Magnuson, K., Huston, A. C., Klebanov, P., Pagani, L. S., Feinstein, L., Engel, M., Brooks-Gunn, J., et al. (2007). School Readiness and Later Achievement. *Developmental Psychology*, 43(6), 1428–1446.

- Firpo, S., Fortin, N. M., & Lemieux, T. (2009). Unconditional Quantile Regressions. *Econometrica*, 77(3), 953–973.
- Fomby, P., & Cherlin, A. J. (2007). Family Instability and Child Well-Being. *American Sociological Review*, 72(2), 181–204.
- Fortin, N., Lemieux, T., & Firpo, S. (2011). Decomposition methods in economics. In *Handbook of labor economics* (pp. 1–102, Vol. 4). Elsevier.
- Fryer, R. G., & Levitt, S. D. (2010). An Empirical Analysis of the Gender Gap in Mathematics. *American Economic Journal: Applied Economics*, 2(2), 210–240.
- Guiso, L., Monte, F., Sapienza, P., & Zingales, L. (2008). Culture, Gender, and Math. *Science*, 320(5880), 1164–1165.
- Hedges, L. V., & Hedberg, E. C. (2007). Intraclass Correlation Values for Planning Group-Randomized Trials in Education. *Educational Evaluation and Policy Analysis*, 29(1), 60–87. <https://doi.org/10.3102/0162373707299706>
- Hyde, J. S., & Mertz, J. E. (2009). Gender, Culture, and Mathematics Performance. *Proceedings of the National Academy of Sciences*, 106(22), 8801–8807.
- Jann, B. (2008). The blinder–oaxaca decomposition for linear regression models. *The Stata Journal*, 8(4), 453–479.
- Juras, R. (2016). Exploring Intraclass Correlation in Random-Intercept Models: The Baseline for Social and Political Indicators. *Journal of School Choice*, 10(4), 574–587. <https://doi.org/10.1080/15582159.2016.1241578>
- Lavy, V., & Sand, E. (2018). On the Origins of Gender Gaps in Human Capital: Short and Long-Term Consequences of Teachers’ Biases. *Journal of Public Economics*, 167, 263–279.
- Lee, D., & McLanahan, S. (2015). Family Structure Transitions and Child Development: Instability, Selection, and Population Heterogeneity. *American Sociological Review*, 80(4), 738–763.
- Lindberg, S. M., Hyde, J. S., Petersen, J. L., & Linn, M. C. (2010). New Trends in Gender and Mathematics Performance: A Meta-Analysis. *Psychological Bulletin*, 136(6), 1123–1135.
- McLanahan, S., Tach, L., & Schneider, D. (2013). The Causal Effects of Father Absence. *Annual Review of Sociology*, 39, 399–427.
- Neumark, D. (1988). Employers’ discriminatory behavior and the estimation of wage discrimination. *Journal of Human Resources*, 23(3), 279–295.
- Nollenberger, N., Rodríguez-Planas, N., & Sevilla, A. (2016). The Math Gender Gap: The Role of Culture. *American Economic Review*, 106(5), 257–61.
- Oaxaca, R. (1973). Male-Female Wage Differentials in Urban Labor Markets. *International Economic Review*, 14(3), 693–709.

- OECD. (2020). *Do Gender Gaps in Reading and Mathematics Evolve Between Childhood and Adulthood?* <https://doi.org/10.1787/6db40c8d-en>
- O'Meara, N., Prendergast, M., & Treacy, P. (2020). What's the Point?: Impact of Ireland's Bonus Points Initiative on Student Profile in Mathematics Classrooms. *Issues in Educational Research*, 30(4), 1418–1441.
- Rios-Avila, F. (2020). Recentered Influence Functions (RIFs) in Stata: RIF Regression and RIF Decomposition. *The Stata Journal*, 20(1), 51–94.
- Sirin, S. R. (2005). Socioeconomic Status and Academic Achievement: A Meta-Analytic Review of Research. *Review of Educational Research*, 75(3), 417–453.
- Spencer, S. J., Steele, C. M., & Quinn, D. M. (1999). Stereotype Threat and Women's Math Performance. *Journal of Experimental Social Psychology*, 35(1), 4–28. <https://doi.org/10.1006/jesp.1998.1373>
- Whitcomb, K. M., Kalender, Z. Y., Nokes-Malach, T. J., Schunn, C. D., & Singh, C. (2020). A Mismatch Between Self-efficacy and Performance: Undergraduate Women in Engineering Tend to Have Lower Self-efficacy Despite Earning Higher Grades Than Men. *International Journal of Engineering Education*, 36(4), 1996–2014.

Appendices

Appendix 6: summary statistics. Appendix 7: OLS baseline regressions (age 9 and age 13 predictors). Appendix 8: family-structure transitions and father-absence identification. Appendices 9–10: variable-level Oaxaca-Blinder decompositions for Maths (gender and father absence). Appendices 11–12: corresponding decompositions for English. Appendix 13: DFL distributional decompositions (Maths); Appendices 14–15: DFL decompositions (English). Appendix 16: school-clustering inference sensitivity. Appendix 17: level-selection sensitivity in Leaving Certificate Maths. Appendix 18: attrition robustness via Lee bounds. Appendix 19: additional robustness checks (harmonised wave comparisons, level-mix decompositions, alternative father-absence definitions). Appendix 20: attrition balance by gender. Appendix 21: Leaving Certificate Maths reporting diagnostics.

6 Appendix A. Summary Statistics

Table 13 reports descriptive statistics for the analytical sample ($n = 4,333$). Leaving Certificate Maths points are harmonised across grading systems and capped at 100. Cognitive skills use standardised logit scores; SDQ subscales range 0–10, with higher scores indicating greater difficulties. The sample comprises cohort members with complete data on the outcome, cognitive assessments, socioemotional indicators, and demographic controls; item-level missingness (not wave-level attrition) is the primary source of exclusion.

Table 14 disaggregates by father-presence status, showing systematic differences in every domain.

Table 13: Summary statistics for the GUI Child Cohort ('98), analytical sample

Variable	<i>N</i>	Mean	SD	Min	Max
<i>Panel A: Leaving Certificate Performance</i>					
Maths Leaving Certificate Points (Raw)	4,333	60.93	32.84	0.00	125.00
Maths Leaving Certificate Points (Harmonised)	4,333	56.03	27.43	0.00	100.00
New Grading System	4,333	0.39	0.49	0.00	1.00
<i>Panel B: Cognitive Skills</i>					
Wave 1 (Age 9)					
Reading Ability (logit)	4,264	0.35	0.66	-3.36	2.87
Maths Ability (logit)	4,306	-0.48	0.61	-3.62	1.90
Wave 2 (Age 13)					
Verbal Reasoning (logit)	4,105	0.15	0.64	-2.37	1.78
Numerical Ability (logit)	4,093	0.14	0.64	-2.36	2.11
BAS Matrix Reasoning (score)	3,939	119.40	9.48	10.00	161.00
<i>Panel C: Non-Cognitive Skills (SDQ)</i>					
Wave 1 (Age 9)					
Emotional Symptoms	4,330	1.94	1.48	0.00	10.00
Conduct Problems	4,328	1.11	0.99	0.00	9.00
Hyperactivity	4,325	2.73	1.49	0.00	10.00
Peer-relationship Problems	4,322	1.07	0.99	0.00	9.00
Wave 2 (Age 13)					
Emotional Symptoms	4,252	1.65	1.48	0.00	10.00
Conduct Problems	4,252	0.95	0.97	0.00	10.00
Hyperactivity	4,252	2.29	0.98	0.00	10.00
Peer-relationship Problems	4,252	1.01	0.98	0.00	10.00
<i>Panel D: Demographic and Family Characteristics</i>					
Male	4,333	0.48	0.50	0.00	1.00
Wave 1 (Age 9)					
Mother's Education: Upper Secondary	4,333	0.56	0.50	0.00	1.00
Mother's Education: Third Level	4,333	0.32	0.47	0.00	1.00
Father's Education: Upper Secondary	3,808	0.46	0.50	0.00	1.00
Father's Education: Third Level	3,808	0.32	0.47	0.00	1.00
Income Quintile	4,033	3.50	1.47	1.00	5.00
Mixed School	4,051	0.76	0.43	0.00	1.00
Father Missing	4,333	0.12	0.33	0.00	1.00
Wave 2 (Age 13)					
Mother's Education: Upper Secondary	4,253	0.56	0.50	0.00	1.00
Mother's Education: Third Level	4,253	0.36	0.48	0.00	1.00
Father's Education: Upper Secondary	3,435	0.49	0.50	0.00	1.00
Father's Education: Third Level	3,435	0.36	0.48	0.00	1.00
Income Quintile	3,960	3.42	1.47	1.00	5.00
Fee-Paying School	4,124	0.11	0.31	0.00	1.00
DEIS School	4,124	0.11	0.31	0.00	1.00
Mixed School	4,023	0.53	0.50	0.00	1.00
Religious School	4,333	0.67	0.47	0.00	1.00
Father Missing	4,333	0.21	0.41	0.00	1.00
Father Consistently Absent	3,700	0.11	0.31	0.00	1.00

Notes: Sample restricted to observations with valid Leaving Certificate Maths outcomes ($N = 4,333$). Continuous variables are reported in original units (points or test-score scales). Binary indicators are reported as proportions. Sample sizes vary across variables due to item-level missingness.

Table 14: Summary statistics by father-presence status

Variable	Father present	Father absent	Difference
<i>Panel A: Leaving Certificate Performance (Wave 4)</i>			
Maths Leaving Certificate Points (Raw)	63.72	44.66	19.06***
Maths Leaving Certificate Points (Harmonised)	58.34	42.49	15.85***
<i>Panel B: Cognitive Skills</i>			
Wave 1 (Age 9)			
Reading Ability (logit)	0.27	-0.08	0.35***
Maths Ability (logit)	-0.54	-0.91	0.37***
Wave 2 (Age 13)			
Verbal Reasoning (logit)	0.06	-0.28	0.34***
Numerical Ability (logit)	0.06	-0.33	0.39***
BAS Matrix Reasoning (score)	117.8	112.4	5.38***
<i>Panel C: Socioemotional Skills (SDQ)</i>			
Wave 1 (Age 9)			
Emotional Symptoms	1.90	2.42	-0.52***
Conduct Problems	1.15	1.56	-0.42***
Hyperactivity	2.83	3.57	-0.74***
Peer Problems	1.03	1.52	-0.49***
Wave 2 (Age 13)			
Emotional Symptoms	1.65	2.21	-0.56***
Conduct Problems	0.99	1.39	-0.40***
Hyperactivity	2.40	3.22	-0.82***
Peer Problems	1.03	1.38	-0.36***
<i>Panel D: Demographic and Family Characteristics</i>			
Wave 1 (Age 9)			
Mother's Education (mean)	3.79	3.35	0.44***
Mother: Upper Secondary	0.57	0.53	0.03**
Mother: Third Level	0.29	0.19	0.10***
Income Quintile	3.54	2.52	1.02***
Mixed School	0.76	0.73	0.03
Wave 2 (Age 13)			
Mother's Education (mean)	3.96	3.62	0.34***
Mother: Upper Secondary	0.57	0.57	0.01
Mother: Third Level	0.33	0.25	0.08***
Income Quintile	3.40	2.72	0.68***
Fee-Paying School	0.11	0.05	0.06***
DEIS School	0.12	0.24	-0.12***
Religious School	0.67	0.43	0.24***
Mixed School	0.53	0.61	-0.08***
Female	0.50	0.57	-0.07***

Notes: Columns report group means by father-presence status. Differences are defined as mean(father present) minus mean(father absent). Continuous variables are reported in original units and binary variables as proportions. Female is coded as an indicator equal to 1 if female. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

7 Appendix B. OLS Regressions for Leaving Certificate Maths

7.1 Model specifications

I estimate four OLS specifications regressing adjusted Leaving Certificate Maths points on cognitive, socioemotional, socioeconomic, and school covariates. These regressions provide context for the decompositions: endowment components are large only when the underlying predictors strongly predict the outcome.

$$\text{Maths Points}_i = \beta_0 + \sum_k \beta_k \text{Cog}_{k,i,w} + \sum_l \beta_l \text{SocEmo}_{l,i,w} + \sum_n \beta_n \text{SES}_{n,i,w} + \sum_x \beta_x \text{School}_{x,i,w} + \varepsilon_i \quad (7.1)$$

Models 1–2 use Wave 1 (age 9) predictors; Models 3–4 use Wave 2 (age 13) predictors. Within each wave, the key variation is the treatment of father’s education: odd-numbered models include the full sample with a father’s-education-missing indicator, while even-numbered models condition on observed father’s education. Table 15 summarises the covariate structure.

Table 15: Variables included in each OLS specification for Leaving Certificate Maths. Checkmarks indicate included covariates. The table documents model design and does not report estimated coefficients.

Variable Group	Model 1	Model 2	Model 3	Model 4
Wave	1 (Age 9)	1 (Age 9)	2 (Age 13)	2 (Age 13)
Cognitive Skills				
Numerical Ability	✓	✓	✓	✓
Reading Ability/Verbal Reasoning	✓	✓	✓	✓
BAS Matrices			✓	✓
Socioemotional Traits				
Emotional Symptoms	✓	✓	✓	✓
Conduct Problems	✓	✓	✓	✓
Hyperactivity	✓	✓	✓	✓
Peer-Relationship Problems	✓	✓	✓	✓
Socioeconomic Status				
Mother’s Education	✓	✓	✓	✓
Father’s Education		✓		✓
Father’s Education Missing	✓		✓	
Income Quintile	✓	✓	✓	✓
Individual & School Factors				
Male	✓	✓	✓	✓
CoEd School	✓	✓	✓	✓
Fee Paying School			✓	✓
DEIS School			✓	✓
Religious School			✓	✓

All models are estimated with heteroskedasticity-robust standard errors. Sample sizes range from 2,777 to 3,690 depending on wave and father’s-education specification. Results are reported

in Tables 16 (Wave 1) and 17 (Wave 2).

7.2 Results

Three patterns emerge from Tables 16 and 17. First, Wave 2 models explain substantially more variance ($R^2 \approx 0.39\text{--}0.40$) than Wave 1 models ($R^2 \approx 0.30$), confirming that later-measured predictors are more informative for an outcome observed at ages 17–18.

Second, cognitive skills dominate. Numerical ability is the strongest predictor at both ages ($\beta \approx 9.6$ at age 9, rising to 11.7 at age 13; $p < 0.001$), consistent with Duncan et al. (2007). Reading/verbal reasoning contributes an additional 3.6–4.7 points. Among socioemotional traits, only hyperactivity is consistently significant ($\beta \approx -1.5$ to -1.8 , $p < 0.001$); conduct problems are significant at age 9 but fade by age 13, while peer problems are never significant. Parental education and income show clear gradients at both ages, and missing father’s education is associated with lower scores (-6.4 points at age 9, -3.0 at age 13), consistent with paternal disengagement effects. School characteristics-fee-paying ($+4.3$ points) and DEIS status (-3.3 to -4.8)-become significant predictors at age 13.

Third, the raw gender coefficient shrinks from significant at age 9 ($\beta \approx 3.4\text{--}3.9$, $p < 0.001$) to insignificant at age 13 ($\beta \approx 0.8\text{--}1.0$). The gender gap itself persists (the O-B decompositions in the main text confirm this); the coefficient shrinks because Wave 2 covariates, especially numerical ability, absorb more of the gap through measured endowments, foreshadowing the compositional shift documented in the main analysis.

Table 16: Predictors of adjusted Leaving Certificate Maths points using Wave 1 covariates

Variable	Model 1	Model 2
Intercept	47.153*** (1.972)	45.770*** (2.149)
Numerical ability	9.637*** (0.587)	9.568*** (0.621)
Reading ability	4.125*** (0.555)	3.612*** (0.588)
Emotional symptoms	-0.441 [†] (0.242)	-0.595* (0.257)
Conduct problems	-0.964** (0.349)	-1.044** (0.371)
Hyperactivity	-1.508*** (0.202)	-1.566*** (0.214)
Peer problems	-0.096 (0.332)	0.055 (0.353)
Mother: upper secondary / technical	5.887*** (1.352)	4.739** (1.507)
Mother: bachelor's / postgraduate	12.808*** (1.524)	10.011*** (1.723)
Father: upper secondary / technical	—	5.886*** (1.167)
Father: bachelor's / postgraduate	—	9.917*** (1.396)
Income quintile	2.459*** (0.352)	1.885*** (0.385)
Male	3.881*** (0.863)	3.350*** (0.907)
Co-educational school	0.957 (0.992)	0.956 (1.041)
Father's education missing	-6.350*** (1.303)	—
Observations	3,690	3,241
Residual standard error	25.28	24.90
Adjusted R^2	0.297	0.299
F statistic	130.6***	107.1***

Notes: Outcome is adjusted Leaving Certificate Maths points (0–100). Model 1 uses the full Wave 1 covariate sample and includes an indicator for missing father's education. Model 2 restricts the sample to observations with observed father's education. Heteroskedasticity-robust standard errors are reported in parentheses. Parental-education coefficients are relative to the omitted category of at most lower secondary education. [†] $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 17: Predictors of adjusted Leaving Certificate Maths points using Wave 2 covariates

Variable	Model 3	Model 4
Intercept	25.557*** (3.813)	24.032*** (4.335)
Numerical ability	11.704*** (0.578)	11.684*** (0.635)
Verbal reasoning	4.688*** (0.566)	4.301*** (0.626)
BAS matrices score	0.199*** (0.026)	0.188*** (0.029)
Emotional symptoms	-0.446 [†] (0.246)	-0.706** (0.273)
Conduct problems	-0.249 (0.367)	-0.432 (0.414)
Hyperactivity	-1.818*** (0.215)	-1.792*** (0.240)
Peer problems	-0.066 (0.317)	-0.114 (0.352)
Mother: upper secondary / technical	3.921* (1.595)	5.003** (1.877)
Mother: bachelor's / postgraduate	7.371*** (1.722)	7.686*** (2.031)
Father: upper secondary / technical	—	2.883* (1.344)
Father: bachelor's / postgraduate	—	5.728*** (1.534)
Income quintile	1.823*** (0.324)	1.469*** (0.372)
Male	0.981 (0.837)	0.834 (0.916)
Fee-paying school	4.287** (1.366)	4.228** (1.478)
DEIS school	-4.802*** (1.423)	-3.310* (1.663)
Religious school	-0.025 (1.082)	0.179 (1.195)
Mixed school	-2.264* (0.979)	-1.639 (1.068)
Father's education missing	-2.961** (1.052)	—
Observations	3,401	2,777
Residual standard error	23.21	23.01
Adjusted R^2	0.397	0.390
F statistic	140.7***	105.5***

Notes: Outcome is adjusted Leaving Certificate Maths points (0–100). Model 3 uses the full Wave 2 covariate sample and includes an indicator for missing father's education. Model 4 restricts the sample to observations with observed father's education. Heteroskedasticity-robust standard errors are reported in parentheses. Parental-education coefficients are relative to the omitted category of at most lower secondary education. [†] $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

8 Appendix C. Family Structure Changes Between Waves

Tables 18–20 trace family-structure transitions across waves and motivate the father-absence proxy used in the main text. Three patterns are especially relevant. First, secondary caregiver (SCG) questionnaire non-completion rises sharply between Waves 2 and 3 (from 5.4% to 11.8% net loss). Second, partnership and marital transitions remain comparatively stable at roughly 3–4% per wave. Third, non-completion increases even among resident SCGs (8.7% to 15.8%), which is difficult to reconcile with a purely mechanical absence story and instead suggests that survey participation captures engagement differences beyond physical absence.

Table 18: Summary of family-dynamics transitions between Wave 1 and Wave 2 (counts and transition percentages conditional on baseline status).

Transition Type	Count	Percentage
Partner Status Transitions		
No partner → No partner	360	85.1% of initially without partner
No partner → Partner	63	14.9% of initially without partner
Partner → Partner	4,141	96.2% of initially with partner
Partner → No partner	165	3.8% of initially with partner
Primary Caregiver Marital Status Transitions		
Married → Married	3,868	95.9% of initially married
Married → Separated	101	2.5% of initially married
Married → Divorced	33	0.8% of initially married
Separated → Separated	118	60.5% of initially separated
Separated → Married	16	8.2% of initially separated
Separated → Divorced	53	27.2% of initially separated
Never married → Never married	291	81.1% of initially never married
Never married → Married	60	16.7% of initially never married
Secondary Caregiver Participation Transitions		
No SCG → No SCG	360	85.1% of initially without SCG
No SCG → SCG (completed)	38	9.0% of initially without SCG
No SCG → SCG (non-completed)	25	5.9% of initially without SCG
SCG (completed) → SCG (completed)	3,619	87.8% of initial SCG completers
SCG (completed) → SCG (non-completed)	359	8.7% of initial SCG completers
SCG (completed) → No SCG	146	3.5% of initial SCG completers
Key Net Changes		
Net partner loss	102	2.4% of initial partnered households
Net increase in separation/divorce	171	4.2% of initial married households
Net decrease in SCG questionnaire completion	223	5.4% of initial SCG completers

Taken together, these transitions support the interpretation of SCG non-response as a meaningful marker of paternal engagement over time and provide the empirical context for the father-absence decompositions in the main analysis.

Table 19: Summary of family-dynamics transitions between Wave 2 and Wave 3 (counts and transition percentages conditional on baseline status).

Transition Type	Count	Percentage
Partner Status Transitions		
No partner → No partner	480	91.6% of initially without partner
No partner → Partner	44	8.4% of initially without partner
Partner → Partner	3,996	95.5% of initially with partner
Partner → No partner	186	4.5% of initially with partner
Primary Caregiver Marital Status Transitions		
Married → Married	3,771	95.1% of initially married
Married → Separated	114	2.9% of initially married
Married → Divorced	11	0.3% of initially married
Separated → Separated	159	65.1% of initially separated
Separated → Married	9	3.7% of initially separated
Separated → Divorced	55	22.5% of initially separated
Divorced → Divorced	85	56.7% of initially divorced
Divorced → Widowed	53	35.3% of initially divorced
Never married → Never married	267	89.0% of initially never married
Never married → Married	32	10.7% of initially never married
Secondary Caregiver Participation Transitions		
No SCG → No SCG	480	91.6% of initially without SCG
No SCG → SCG (completed)	17	3.2% of initially without SCG
No SCG → SCG (non-completed)	27	5.2% of initially without SCG
SCG (completed) → SCG (completed)	3,009	80.4% of initial SCG completers
SCG (completed) → SCG (non-completed)	584	15.6% of initial SCG completers
SCG (completed) → No SCG	149	4.0% of initial SCG completers
Key Net Changes		
Net partner loss	142	3.4% of initial partnered households
Net increase in separation/divorce	170	4.3% of initial married households
Net decrease in SCG questionnaire completion	436	11.8% of initial SCG completers

Table 20: Comparison of family-dynamics transitions between Waves 1-2 and 2-3 (counts with within-transition percentages in parentheses).

Transition Type	Wave 1 → Wave 2	Wave 2 → Wave 3
Partner Status Transitions		
No partner → No partner	360 (85.1%)	480 (91.6%)
No partner → Partner	63 (14.9%)	44 (8.4%)
Partner → Partner	4,141 (96.2%)	3,996 (95.5%)
Partner → No partner	165 (3.8%)	186 (4.5%)
Primary Caregiver Marital Status Transitions		
Married → Married	3,868 (95.9%)	3,771 (95.1%)
Married → Separated	101 (2.5%)	114 (2.9%)
Married → Divorced	33 (0.8%)	11 (0.3%)
Separated → Separated	118 (60.5%)	159 (65.1%)
Separated → Married	16 (8.2%)	9 (3.7%)
Separated → Divorced	53 (27.2%)	55 (22.5%)
Never married → Never married	291 (81.1%)	267 (89.0%)
Never married → Married	60 (16.7%)	32 (10.7%)
Secondary Caregiver Participation Transitions		
No SCG → No SCG	360 (85.1%)	480 (91.6%)
No SCG → SCG (completed)	38 (9.0%)	17 (3.2%)
No SCG → SCG (non-completed)	25 (5.9%)	27 (5.2%)
SCG (completed) → SCG (completed)	3,619 (87.8%)	3,009 (80.4%)
SCG (completed) → SCG (non-completed)	359 (8.7%)	584 (15.6%)
SCG (completed) → No SCG	146 (3.5%)	149 (4.0%)
Key Net Changes		
Net partner loss	102 (2.4%)	142 (3.4%)
Net increase in separation/divorce	171 (4.2%)	170 (4.3%)
Net decrease in SCG questionnaire completion	223 (5.4%)	436 (11.8%)

Table 21: Wave 3 validation split of the father-absence proxy

Group	Wave 3 validation category	<i>n</i>	Share (%)	Mean Leaving Certificate Maths	SD
Father-absent	No resident partner (structural absence)	233	61.2	45.92	34.11
Father-absent	Missing Wave 3 partner status	78	20.5	44.08	33.68
Father-absent	Resident partner without SCG completion (disengaged)	47	12.3	44.06	32.79
Father-absent	Resident partner with SCG completion (engaged)	23	6.0	52.09	35.09
Father-present	Resident partner with SCG completion (engaged)	2,040	61.7	64.78	35.39
Father-present	Missing Wave 3 partner status	795	24.1	65.58	37.01
Father-present	Resident partner without SCG completion (disengaged)	380	11.5	59.19	35.10
Father-present	No resident partner (structural absence)	90	2.7	52.21	34.19

Notes: Shares are reported within the father-present and father-absent groups. The table uses Wave 3 household partner status and SCG completion to distinguish structural absence (no resident partner) from resident-partner non-completion of the SCG questionnaire. The 61.2% structural-absence share for the father-absent group is conditional on having non-missing Wave 3 partner information; 20.5% of the father-absent group has missing Wave 3 validation, with Leaving Certificate Maths means close to the structural-absence and resident-disengaged categories.

9 Appendix D. Oaxaca Decompositions: Gender Gaps in Maths, Leaving Cert

This appendix reports the variable-level Oaxaca-Blinder decompositions behind the main-text gender figures. The same pattern shown in the headline plots is visible at the detailed level: with age 9 predictors, the gap is primarily coefficient-led, while with age 13 predictors measured skill differences, especially numeracy, take the larger share.

Table 22: Oaxaca–Blinder decomposition of adjusted Leaving Certificate Maths points using Wave 1 predictors

Statistic	No Father		With Father	
Group 1 (Girls)	52.831***	(0.688)	54.749***	(0.717)
Group 2 (Boys)	58.043***	(0.732)	59.183***	(0.729)
Difference	-5.212***	(0.980)	-4.434***	(1.020)
Endowments	-1.570**	(0.610)	-1.298**	(0.656)
Coefficients	-4.215***	(0.900)	-3.641***	(0.924)
Interaction	0.572	(0.490)	0.505	(0.502)

Variable	Endowments		Coefficients		Interactions	
	No Father	With Father	No Father	With Father	No Father	With Father
Reading Ability	0.021 (0.101)	0.033 (0.090)	0.659* (0.350)	0.722* (0.401)	0.014 (0.070)	0.028 (0.090)
Maths Ability	-1.886*** (0.358)	-1.762*** (0.374)	0.864* (0.476)	0.809* (0.479)	0.376* (0.223)	0.346 (0.225)
Emotional Symptoms	-0.004 (0.109)	-0.044 (0.096)	-1.343 (0.856)	-1.333 (0.902)	-0.208 (0.151)	-0.191 (0.143)
Conduct Problems	0.262** (0.120)	0.327*** (0.123)	1.354 (0.927)	1.595* (0.839)	-0.194 (0.150)	-0.259 (0.166)
Hyperactivity	0.784*** (0.195)	0.857*** (0.191)	-1.645 (1.298)	-1.069 (1.331)	0.340 (0.275)	0.224 (0.287)
Peer-relationship Problems	0.000 (0.026)	0.003 (0.027)	-0.250 (0.690)	-0.320 (0.759)	-0.003 (0.031)	-0.004 (0.040)
Mothers Educ. (Higher 2ndary/Tech)	0.007 (0.097)	0.003 (0.054)	0.548 (1.574)	2.069 (1.762)	0.001 (0.046)	0.004 (0.082)
Mothers Educ. (Bachelor's/Postgrad)	-0.353* (0.194)	-0.174 (0.133)	0.958 (1.106)	1.211 (1.320)	-0.090 (0.133)	-0.079 (0.115)
Fathers Educ. (Higher 2ndary/Tech)	- (0.098)	0.058 (0.098)	- (0.990)	-0.197 (0.990)	- (0.990)	-0.004 (0.036)
Fathers Educ. (Bachelor's/Postgrad)	- (0.169)	-0.488*** (0.169)	- (1.011)	-0.246 (1.011)	- (1.011)	0.035 (0.149)
Father's Educ. Missing	-0.131* (0.072)	- (0.072)	-0.140 (0.303)	- (0.303)	-0.030 (0.074)	- (0.074)
Income Quintile	-0.227* (0.125)	-0.062 (0.079)	0.629 (2.642)	2.067 (2.965)	-0.017 (0.082)	-0.022 (0.061)
Mixed School	-0.043 (0.150)	-0.049 (0.159)	2.228* (1.332)	2.359 (1.520)	0.384 (0.235)	0.426 (0.278)
Constant	-	-	-8.076* (4.210)	-11.309** (4.881)	-	-

The outcome is adjusted Leaving Certificate Maths points on a 0–100 scale. The decomposition uses Wave 1 (age 9) predictors and boys' coefficients as the reference group. “No Father” uses the full sample and includes an indicator for missing father's education ($n = 3,690$; girls 1,886, boys 1,804). “With Father” restricts the sample to cases with observed father's education ($n = 3,241$; girls 1,635, boys 1,606). Endowments capture differences in observed characteristics, coefficients capture differences in returns to those characteristics, and interaction captures their joint contribution. Negative values indicate components associated with higher male scores. Bootstrap standard errors are based on 1,000 replications. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 23: Oaxaca–Blinder decomposition of adjusted Leaving Certificate Maths points using Wave 2 predictors

Statistic	No Father		With Father			
Group 1 (Girls)	54.209***	(0.699)	56.272***	(0.726)		
Group 2 (Boys)	59.091***	(0.732)	60.907***	(0.892)		
Difference	-4.882***	(0.998)	-4.635***	(1.150)		
Endowments	-4.150***	(0.742)	-4.039***	(0.822)		
Coefficients	-1.236	(0.855)	-1.128	(0.963)		
Interaction	0.504	(0.531)	0.532	(0.573)		

Variable	Endowments		Coefficients		Interactions	
	No Father	With Father	No Father	With Father	No Father	With Father
Verbal Reasoning	-1.034*** (0.214)	-0.803*** (0.232)	-0.389 (0.268)	-0.324 (0.392)	0.275 (0.194)	0.183 (0.223)
Numerical Ability	-3.658*** (0.405)	-3.546*** (0.454)	-0.151 (0.365)	-0.325 (0.405)	0.146 (0.349)	0.266 (0.326)
Matrices	-0.186 (0.142)	-0.259 (0.155)	-0.025 (0.039)	-0.109 (0.078)	0.061 (0.066)	0.115 (0.099)
Emotional Symptoms	-0.006 (0.124)	-0.083 (0.161)	-1.130 (0.692)	-1.143 (0.797)	-0.269 (0.182)	-0.299 (0.237)
Conduct Problems	0.012 (0.041)	0.023 (0.055)	0.316 (0.801)	0.219 (0.795)	-0.009 (0.048)	-0.009 (0.051)
Hyperactivity	0.970*** (0.229)	0.948*** (0.222)	-1.636 (1.280)	-1.303 (1.189)	0.401 (0.323)	0.322 (0.296)
Peer-relationship Problems	0.080 (0.064)	0.059 (0.067)	1.124 (0.765)	0.601 (0.712)	-0.144 (0.113)	-0.083 (0.095)
Mothers Educ. (Higher 2ndary/Tech)	0.099 (0.094)	0.100 (0.133)	-1.010 (1.552)	1.211 (2.100)	-0.038 (0.079)	0.056 (0.141)
Mothers Educ. (Bachelor's/Postgrad)	-0.217 (0.137)	-0.179 (0.161)	0.863 (1.295)	1.847 (1.617)	-0.076 (0.143)	-0.151 (0.183)
Fathers Educ. (Higher 2ndary/Tech)	- (0.093)	0.081 (0.093)	- (0.093)	0.570 (1.224)	- (0.093)	0.045 (0.101)
Fathers Educ. (Bachelor's/Postgrad)	- (0.148)	-0.277* (0.148)	- (0.148)	0.485 (1.223)	- (0.148)	-0.068 (0.183)
Father Educ. Missing	-0.123 (0.082)	- (0.082)	0.101 (0.360)	- (0.360)	0.022 (0.084)	- (0.084)
Income Quintile	-0.063 (0.088)	-0.002 (0.065)	0.635 (2.018)	1.532 (2.601)	-0.007 (0.034)	-0.001 (0.044)
Fee Paying School	-0.126 (0.083)	-0.142* (0.069)	0.048 (0.322)	-0.100 (0.315)	-0.012 (0.088)	0.024 (0.076)
Other School Variables†	-	-	-	-	-	-
Constant	-	-	1.277 (4.424)	-3.074 (5.072)	-	-

The outcome is adjusted Leaving Certificate Maths points on a 0–100 scale. The decomposition uses Wave 2 (age 13) predictors and boys' coefficients as the reference group. "No Father" uses the full sample and includes an indicator for missing father's education ($n = 3,401$; girls 1,724, boys 1,677). "With Father" restricts the sample to cases with observed father's education ($n = 2,777$; girls 1,377, boys 1,400). Endowments capture differences in observed characteristics, coefficients capture differences in returns to those characteristics, and interaction captures their joint contribution. Negative values indicate components associated with higher male scores. Bootstrap standard errors are based on 1,000 replications. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Other school variables were included but are not shown because they were not statistically significant.

10 Appendix E. Oaxaca Decompositions: Father Absence Effects in Maths, Leaving Cert

The detailed decomposition results underlying the figures on father absence effects in the main text are reported in the tables below.

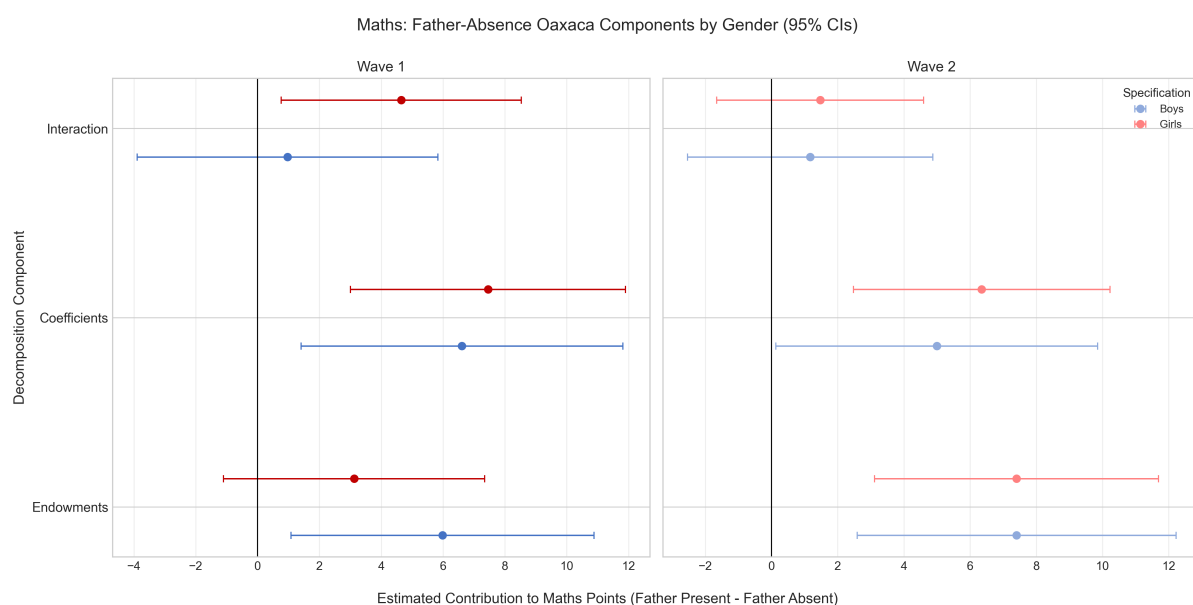


Figure 4: Coefficient-plot representation of Oaxaca-Blinder components for father-absence gaps in Leaving Certificate Maths, estimated separately by gender. Points denote component estimates (endowments, coefficients, interaction) and horizontal whiskers denote 95% confidence intervals based on bootstrap standard errors (1,000 replications). Panels separate Wave 1 (age 9 predictors) and Wave 2 (age 13 predictors). Positive values indicate components associated with higher scores for the father-present group relative to the father-absent group. Estimates are descriptive and should not be interpreted as causal effects.

Figure 4 complements Tables 24 and 25 by presenting the same components as point estimates with 95% confidence intervals, making the cross-wave and cross-gender uncertainty comparisons more transparent.

Table 24: Oaxaca–Blinder decomposition of adjusted Leaving Certificate Maths points by father-presence status using Wave 1 predictors

Statistic	Boys		Girls	
Group 1 (Father Present)	60.834***	(0.892)	55.799***	(0.846)
Group 2 (Father Absent)	47.270***	(2.759)	40.573***	(2.247)
Difference	13.564***	(2.921)	15.225***	(2.415)
Endowments	5.984**	(2.502)	3.122	(2.156)
Coefficients	6.612**	(2.655)	7.456***	(2.270)
Interaction	0.969	(2.483)	4.647**	(1.982)

Variable	Endowments		Coefficients		Interactions	
	Boys	Girls	Boys	Girls	Boys	Girls
Reading Ability	0.842 (0.790)	0.294 (0.572)	-0.285 (1.078)	-0.378 (1.033)	-0.238 (0.771)	-0.058 (0.303)
Maths Ability	2.807** (1.271)	1.326 (0.900)	0.193 (2.410)	-1.425 (2.397)	-0.083 (1.053)	0.391 (0.639)
Emotional Symptoms	-0.929 (0.858)	0.252 (0.561)	-3.883 (3.164)	-2.351 (2.583)	0.837 (0.862)	0.513 (0.634)
Conduct Problems	-0.379 (0.557)	-0.048 (0.537)	-5.076* (2.925)	-0.156 (2.316)	0.872 (0.744)	0.037 (0.562)
Hyperactivity	1.177 (1.221)	0.634 (0.651)	-0.081 (4.776)	-2.602 (3.287)	0.020 (1.268)	0.557 (0.788)
Peer-relationship Problems	1.049 (0.794)	0.001 (0.424)	3.358 (2.399)	0.528 (2.066)	-1.181 (0.898)	-0.123 (0.470)
Mothers Educ. (Higher 2ndary/Tech)	0.569 (1.041)	-0.183 (0.358)	-9.889*** (3.570)	5.997* (3.074)	-0.414 (0.771)	0.527 (0.625)
Mothers Educ. (Bachelor's/Postgrad)	0.944 (1.111)	1.290 (1.154)	-3.294 (2.411)	-1.028 (1.879)	-0.424 (0.661)	-0.283 (0.809)
Income Quintile	-0.038 (1.422)	-0.616 (1.639)	6.929 (6.580)	9.781* (5.141)	1.523 (1.589)	3.180* (1.810)
Mixed School	-0.058 (0.407)	0.172 (0.292)	-4.745 (4.288)	-1.262 (3.969)	0.057 (0.423)	-0.092 (0.306)
Constant	—	—	23.384** (9.465)	0.352 (8.197)	—	—

The outcome is adjusted Leaving Certificate Maths points on a 0–100 scale. The decomposition uses Wave 1 (age 9) predictors and is estimated separately for boys ($n = 1,314$; father present 1,188, father absent 126) and girls ($n = 1,292$; father present 1,142, father absent 150), using father-present coefficients as the reference group. Endowments capture differences in observed characteristics, coefficients capture differences in returns to those characteristics, and interaction captures their joint contribution. Positive values indicate components associated with higher scores among father-present students. Bootstrap standard errors are based on 1,000 replications. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 25: Oaxaca–Blinder decomposition of adjusted Leaving Certificate Maths points by father-presence status using Wave 2 predictors

Statistic	Boys		Girls	
Group 1 (Father Present)	60.834***	(0.844)	55.799***	(0.850)
Group 2 (Father Absent)	47.270***	(2.752)	40.573***	(2.370)
Difference	13.564***	(2.875)	15.225***	(2.520)
Endowments	7.406***	(2.456)	7.405***	(2.189)
Coefficients	4.992**	(2.482)	6.351***	(1.977)
Interaction	1.166	(1.891)	1.469	(1.594)

Variable	Endowments		Coefficients		Interactions	
	Boys	Girls	Boys	Girls	Boys	Girls
Verbal Reasoning	1.410 (1.008)	1.165* (0.639)	-0.945 (0.989)	0.169 (0.326)	-0.742 (0.651)	-0.396 (0.576)
Numerical Ability	2.020 (1.552)	2.940** (1.266)	0.372 (0.710)	-0.428 (0.941)	1.490 (1.330)	0.511 (1.114)
BAS Matrices	0.880 (0.775)	0.638 (0.449)	-0.112 (0.523)	0.201 (0.370)	0.151 (0.685)	-0.177 (0.350)
Emotional Symptoms	-0.608 (0.662)	0.160 (0.502)	-2.156 (2.553)	-2.202 (2.518)	0.593 (0.745)	0.440 (0.609)
Conduct Problems	-0.046 (0.244)	0.074 (0.318)	-1.420 (1.767)	0.226 (1.897)	0.108 (0.342)	-0.037 (0.326)
Hyperactivity	1.504 (0.921)	1.458* (0.825)	-0.336 (3.288)	3.418 (3.294)	0.099 (0.986)	-0.658 (0.693)
Peer-relationship Problems	0.861 (0.666)	-0.851 (0.595)	3.349 (2.365)	-2.479 (1.872)	-0.767 (0.708)	0.752 (0.607)
Mothers Educ. (Higher 2ndary/Tech)	0.411 (0.767)	-0.114 (0.508)	-4.969 (3.491)	8.428*** (3.120)	-0.277 (0.573)	0.191 (0.790)
Mothers Educ. (Bachelor's/Postgrad)	0.235 (0.502)	0.491 (0.818)	0.001 (3.044)	1.454 (2.655)	0.000 (0.403)	0.376 (0.799)
Income Quintile	0.967 (0.706)	0.151 (1.047)	-3.413 (5.682)	5.468 (5.425)	-0.421 (0.702)	1.183 (1.126)
Fee Paying School	-1.065* (0.619)	0.371 (0.756)	1.884** (0.769)	-0.210 (0.519)	1.246* (0.695)	-0.176 (0.740)
Constant	–	–	18.589 (11.002)	-10.182 (10.764)	–	–

The outcome is adjusted Leaving Certificate Maths points on a 0–100 scale. The decomposition uses Wave 2 (age 13) predictors and is estimated separately for boys ($n = 1,314$) and girls ($n = 1,292$), using father-present coefficients as the reference group. Endowments capture differences in observed characteristics, coefficients capture differences in returns to those characteristics, and interaction captures their joint contribution. Positive values indicate components associated with higher scores among father-present students. Bootstrap standard errors are based on 1,000 replications. Contributions from other school control variables (DEIS School, Mixed School, and Religious School) are included in the model but are not itemised separately; they are incorporated into the Endowments, Coefficients, and Interaction totals. This is why the sum of the itemised rows does not exactly match the component totals. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

11 Appendix F. Oaxaca Decompositions: Gender Gaps in English, Leaving Cert

This appendix reports the variable-level Oaxaca-Blinder decompositions for the English gender gap (Tables 26 and 27), which provide the cross-subject comparison discussed in the main text.

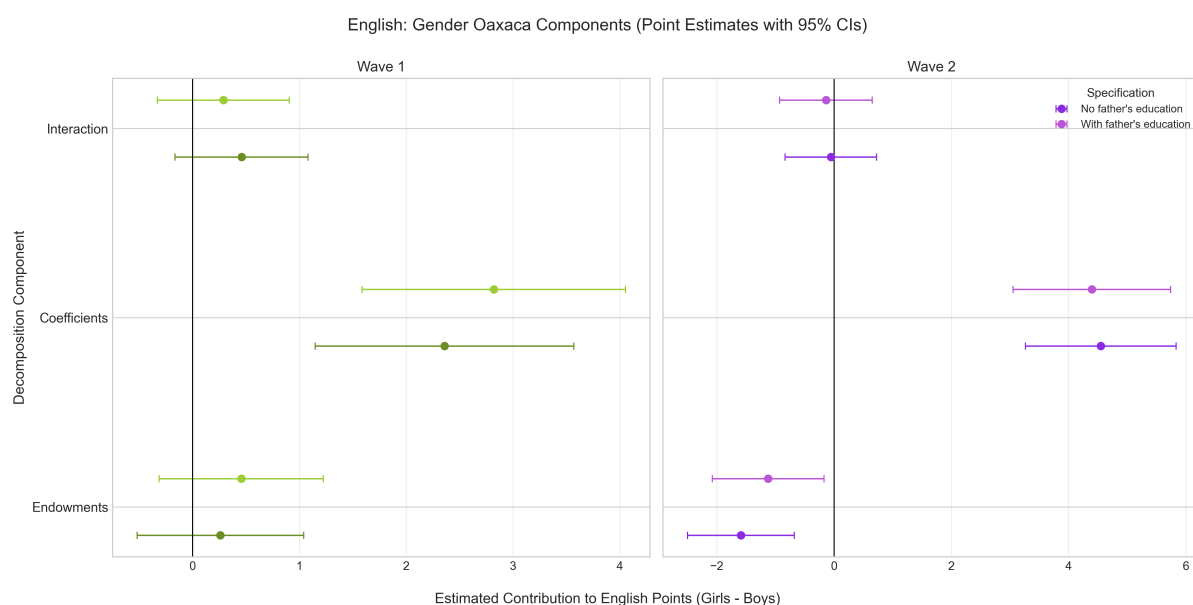


Figure 5: Coefficient-plot representation of Oaxaca-Blinder components for the gender gap in Leaving Certificate English. Points denote component estimates (endowments, coefficients, interaction) and horizontal whiskers denote 95% confidence intervals based on bootstrap standard errors (1,000 replications). Panels separate Wave 1 (age 9 predictors) and Wave 2 (age 13 predictors), each shown with and without father's education controls. Positive values indicate components associated with higher female scores. Estimates are descriptive and should not be interpreted as causal effects.

Figure 5 presents the same decomposition quantities as the appendix tables in coefficient-plot form, which facilitates direct inspection of uncertainty around the reversal pattern between endowments and coefficients in English.

Across models, the decomposition results show that girls score higher than boys in English at the Leaving Certificate, with an average gap of about 3 points across both the Wave 1 and Wave 2 models. Most of this gap comes from the coefficients component-meaning that, for the same set of observed characteristics, the mapping into outcomes differs between boys and girls. Differences in measured skill levels themselves explain a smaller part of the gap. The English gap therefore reads as coefficient-dominant, with the same broad pattern across models with and without paternal education.

Table 26: Oaxaca–Blinder decomposition of adjusted Leaving Certificate English points using Wave 1 predictors

Statistic	No Father		With Father	
Group 1 (Girls)	69.065***	(0.434)	69.875***	(0.444)
Group 2 (Boys)	65.986***	(0.492)	66.309***	(0.497)
Difference	3.079***	(0.649)	3.566***	(0.677)
Endowments	0.260	(0.398)	0.456	(0.393)
Coefficients	2.360***	(0.618)	2.821***	(0.630)
Interaction	0.459	(0.318)	0.288	(0.315)

Variable	Endowments		Coefficients		Interactions	
	No Father	With Father	No Father	With Father	No Father	With Father
Reading Ability	0.079 (0.223)	0.146 (0.219)	0.015 (0.260)	-0.056 (0.268)	0.001 (0.024)	-0.003 (0.027)
Maths Ability	-0.190* (0.112)	-0.147 (0.101)	0.173 (0.315)	0.129 (0.291)	0.074 (0.132)	0.053 (0.122)
Emotional Symptoms	0.034 (0.077)	0.063 (0.075)	0.146 (0.627)	-0.266 (0.657)	0.022 (0.096)	-0.038 (0.096)
Conduct Problems	0.135** (0.068)	0.178** (0.084)	0.290 (0.566)	0.614 (0.602)	-0.044 (0.089)	-0.104 (0.106)
Hyperactivity	0.688*** (0.158)	0.705*** (0.173)	-0.207 (0.853)	-0.051 (0.946)	0.043 (0.178)	0.011 (0.208)
Peer-relationship Problems	0.000 (0.017)	-0.001 (0.018)	-0.510 (0.527)	-0.352 (0.557)	-0.010 (0.033)	-0.006 (0.038)
Mothers Educ. (Higher 2ndary/Tech)	0.004 (0.055)	0.001 (0.055)	0.947 (1.237)	0.911 (1.258)	0.002 (0.050)	0.000 (0.052)
Mothers Educ. (Bachelor's/Postgrad)	-0.196* (0.113)	-0.098 (0.087)	0.703 (0.791)	0.751 (0.816)	-0.065 (0.095)	-0.045 (0.079)
Fathers Educ. (Higher 2ndary/Tech)	- (0.059)	0.019 (0.059)	- (0.724)	0.417 (0.724)	- (0.037)	0.006 (0.037)
Fathers Educ. (Bachelor's/Postgrad)	- (0.106)	-0.178* (0.106)	- (0.664)	0.635 (0.664)	- (0.097)	-0.087 (0.097)
Father's Educ. Missing	-0.019 (0.044)	- (0.044)	-0.166 (0.218)	- (0.218)	-0.043 (0.062)	- (0.062)
Income Quintile	-0.070 (0.054)	-0.013 (0.029)	0.803 (1.885)	1.072 (1.871)	-0.024 (0.060)	-0.012 (0.036)
Mixed School	-0.205* (0.115)	-0.217** (0.110)	2.944** (0.972)	2.803** (0.917)	0.502*** (0.181)	0.512*** (0.176)
Constant	-	-	-2.777 (2.658)	-3.785 (2.778)	-	-

The outcome is adjusted Leaving Certificate English points. The decomposition uses Wave 1 (age 9) predictors and boys' coefficients as the reference group. "No Father" uses the full sample and includes an indicator for missing father's education ($n = 3,679$; girls 1,887, boys 1,792). "With Father" restricts the sample to cases with observed father's education ($n = 3,233$; girls 1,633, boys 1,600). Endowments capture differences in observed characteristics, coefficients capture differences in returns to those characteristics, and interaction captures their joint contribution. Positive values indicate components associated with higher female scores. Bootstrap standard errors are based on 1,000 replications. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 27: Oaxaca–Blinder decomposition of adjusted Leaving Certificate English points using Wave 2 predictors

Statistic	No Father		With Father	
Group 1 (Girls)	69.557***	(0.483)	70.585***	(0.501)
Group 2 (Boys)	66.650***	(0.457)	67.450***	(0.630)
Difference	2.906***	(0.678)	3.135***	(0.726)
Endowments	-1.588***	(0.465)	-1.123**	(0.486)
Coefficients	4.548***	(0.655)	4.396***	(0.685)
Interaction	-0.054	(0.399)	-0.138	(0.403)

Variable	Endowments		Coefficients		Interactions	
	No Father	With Father	No Father	With Father	No Father	With Father
Verbal Reasoning	-1.655*** (0.253)	-1.363*** (0.290)	-0.411 (0.236)	-0.577* (0.303)	0.282* (0.163)	0.312* (0.185)
Numerical Ability	-0.553*** (0.211)	-0.420** (0.193)	0.235 (0.288)	0.420 (0.309)	-0.228 (0.281)	-0.343 (0.259)
Matrices	-0.026 (0.036)	-0.041 (0.046)	-0.018 (0.030)	-0.059 (0.051)	0.047 (0.052)	0.061 (0.062)
Emotional Symptoms	-0.013 (0.089)	0.016 (0.110)	-0.289 (0.421)	-0.388 (0.537)	-0.069 (0.103)	-0.105 (0.149)
Conduct Problems	-0.011 (0.025)	-0.003 (0.027)	-0.693 (0.575)	-0.539 (0.555)	0.022 (0.042)	0.020 (0.042)
Hyperactivity	0.677*** (0.169)	0.720*** (0.173)	-0.107 (0.931)	0.008 (0.888)	0.026 (0.226)	-0.002 (0.215)
Peer-relationship Problems	0.043 (0.044)	0.026 (0.044)	-0.115 (0.496)	-0.298 (0.525)	0.014 (0.065)	0.038 (0.075)
Mothers Educ. (Higher 2ndary/Tech)	0.082 (0.089)	0.086 (0.077)	-1.196 (1.507)	-0.854 (1.680)	-0.037 (0.091)	-0.036 (0.090)
Mothers Educ. (Bachelor's/Postgrad)	-0.178 (0.119)	-0.124 (0.096)	-0.614 (1.136)	-0.240 (1.313)	0.052 (0.115)	0.018 (0.115)
Fathers Educ. (Higher 2ndary/Tech)	” (0.056)	0.048 (0.056)	” (0.910)	1.001 (0.910)	” (0.095)	0.082 (0.095)
Fathers Educ. (Bachelor's/Postgrad)	” (0.089)	-0.111 (0.089)	” (0.900)	1.051 (0.900)	” (0.136)	-0.153 (0.136)
Father Educ. Missing	-0.053 (0.053)	” (0.061)	0.121 (0.246)	” (0.061)	0.030 (0.061)	” (0.061)
Income Quintile	-0.027 (0.042)	-0.002 (0.024)	1.928 (1.470)	2.205 (2.148)	-0.022 (0.035)	-0.003 (0.049)
Mixed School	0.136* (0.071)	0.075 (0.064)	1.406* (0.737)	0.828 (0.819)	-0.159 (0.102)	-0.076 (0.085)
Constant	”	”	2.888 (3.371)	0.817 (3.982)	”	”

The outcome is adjusted Leaving Certificate English points. The decomposition uses Wave 2 (age 13) predictors and boys' coefficients as the reference group. “No Father” uses the full sample and includes an indicator for missing father's education ($n = 3,375$; girls 1,719, boys 1,656). “With Father” restricts the sample to cases with observed father's education ($n = 2,756$; girls 1,370, boys 1,386). Endowments capture differences in observed characteristics, coefficients capture differences in returns to those characteristics, and interaction captures their joint contribution. Positive values indicate components associated with higher female scores. Contributions from school control variables (DEIS, fee-paying, mixed, and religious school) are included in the model but not itemised separately; they are incorporated into the component totals, so the sum of the itemised rows may not exactly match. Bootstrap standard errors are based on 1,000 replications. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

12 Appendix G. Oaxaca Decompositions: Father Absence Effects in English, Leaving Cert

This appendix reports Oaxaca-Blinder decompositions for father-absence gaps in English, complementing the Maths results in Appendix 10. The broad pattern is similar across subjects: father absence is associated with lower achievement for both boys and girls, and much of that association is captured by endowment differences, especially cognitive traits such as verbal reasoning. Coefficient effects are smaller and less stable, and interaction terms are generally imprecise. This is consistent with the interpretation in the main text that measured-skill differences account for most of the father-absence penalty.

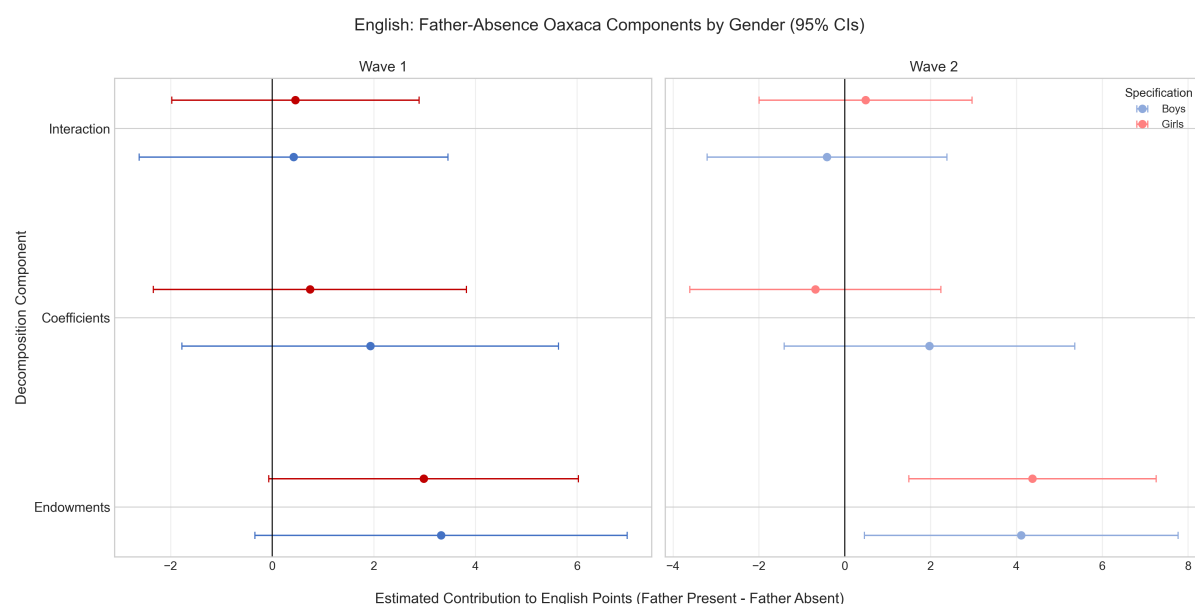


Figure 6: Coefficient-plot representation of Oaxaca-Blinder components for father-absence gaps in Leaving Certificate English, estimated separately by gender. Points denote component estimates (endowments, coefficients, interaction) and horizontal whiskers denote 95% confidence intervals based on bootstrap standard errors (1,000 replications). Panels separate Wave 1 (age 9 predictors) and Wave 2 (age 13 predictors). Positive values indicate components associated with higher scores for the father-present group relative to the father-absent group. Estimates are descriptive and should not be interpreted as causal effects.

Figure 6 provides a compact visual summary of the English father-absence decompositions in Tables 28 and 29, with confidence intervals that highlight which component differences are estimated with greater precision.

Table 28: Oaxaca–Blinder decomposition of adjusted Leaving Certificate English points by father-presence status using Wave 1 predictors

Statistic	Boys		Girls	
Group 1 (Father Present)	66.911***	(0.568)	70.569***	(0.544)
Group 2 (Father Absent)	61.238***	(2.023)	66.391***	(1.634)
Difference	5.673***	(2.105)	4.179**	(1.707)
Endowments	3.323*	(1.869)	2.980*	(1.554)
Coefficients	1.930	(1.890)	0.742	(1.572)
Interaction	0.420	(1.551)	0.457	(1.242)

Variable	Endowments		Coefficients		Interactions	
	Boys	Girls	Boys	Girls	Boys	Girls
Reading Ability	1.670 (1.067)	0.622 (0.700)	-0.278 (0.490)	-0.198 (0.616)	-0.243 (0.592)	-0.049 (0.223)
Maths Ability	0.222 (0.597)	0.131 (0.354)	0.273 (1.403)	0.002 (1.361)	-0.110 (0.613)	-0.000 (0.387)
Emotional Symptoms	0.115 (0.528)	-0.457 (0.460)	0.833 (2.463)	-2.162 (2.219)	-0.181 (0.560)	0.486 (0.517)
Conduct Problems	-0.266 (0.384)	0.015 (0.393)	-2.634 (1.730)	0.732 (1.799)	0.484 (0.466)	-0.167 (0.437)
Hyperactivity	0.707 (0.810)	0.729 (0.490)	-1.826 (3.087)	-1.135 (2.232)	0.465 (0.814)	0.235 (0.544)
Peer-relationship Problems	-0.192 (0.614)	0.335 (0.348)	-0.795 (1.746)	0.794 (1.438)	0.282 (0.651)	-0.197 (0.360)
Mothers Educ. (Higher 2ndary/Tech)	0.263 (0.585)	0.044 (0.280)	-2.557 (3.364)	2.145 (2.431)	-0.140 (0.493)	0.139 (0.315)
Mothers Educ. (Bachelor's/Postgrad)	0.833 (1.065)	0.222 (0.366)	-4.867** (2.302)	1.362 (1.270)	-0.598 (0.820)	0.398 (0.477)
Income Quintile	-0.073 (0.985)	1.148 (0.939)	2.067 (4.838)	-0.758 (2.900)	0.457 (1.060)	-0.256 (0.951)
Mixed School	0.044 (0.218)	0.192 (0.250)	-0.126 (3.290)	-2.134 (3.107)	0.005 (0.220)	-0.131 (0.253)
Constant	–	–	11.839 (7.898)	2.095 (6.873)	–	–

The outcome is adjusted Leaving Certificate English points. The decomposition uses Wave 1 (age 9) predictors and is estimated separately for boys ($n = 1,300$; father present 1,178, father absent 122) and girls ($n = 1,289$; father present 1,138, father absent 151), using father-present coefficients as the reference group. Endowments capture differences in observed characteristics, coefficients capture differences in returns to those characteristics, and interaction captures their joint contribution. Positive values indicate components associated with higher scores among father-present students. Bootstrap standard errors are based on 1,000 replications. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 29: Oaxaca–Blinder decomposition of adjusted Leaving Certificate English points by father-presence status using Wave 2 predictors

Statistic	Boys		Girls	
Group 1 (Father Present)	66.911***	(0.585)	70.569***	(0.539)
Group 2 (Father Absent)	61.238***	(1.944)	66.391***	(1.591)
Difference	5.673***	(2.036)	4.179**	(1.672)
Endowments	4.112**	(1.864)	4.376***	(1.469)
Coefficients	1.975	(1.728)	-0.684	(1.492)
Interaction	-0.413	(1.425)	0.487	(1.267)

Variable	Endowments		Coefficients		Interactions	
	Boys	Girls	Boys	Girls	Boys	Girls
Verbal Reasoning	1.450 (1.212)	1.918*** (0.706)	-0.925 (0.800)	0.207 (0.251)	-0.534 (0.616)	-0.524 (0.430)
Numerical Ability	-1.031 (0.935)	1.056 (0.709)	0.451 (0.500)	0.257 (0.538)	1.330 (0.951)	-0.308 (0.647)
BAS Matrices	0.488 (0.480)	-0.108 (0.484)	0.165 (0.339)	-0.013 (0.524)	-0.231 (0.443)	0.012 (0.494)
Emotional Symptoms	0.169 (0.515)	0.209 (0.455)	0.839 (1.898)	0.534 (2.185)	-0.248 (0.572)	-0.095 (0.453)
Conduct Problems	-0.107 (0.367)	-0.010 (0.242)	-0.822 (2.007)	-0.126 (1.454)	0.104 (0.373)	0.018 (0.249)
Hyperactivity	1.218 (0.892)	0.435 (0.516)	-0.627 (2.953)	-0.225 (2.693)	0.191 (0.903)	0.039 (0.519)
Peer-relationship Problems	-0.199 (0.304)	-0.035 (0.592)	-1.076 (1.360)	-0.793 (1.935)	0.251 (0.355)	0.237 (0.653)
Mothers Educ. (Higher 2ndary/Tech)	0.142 (0.410)	-0.031 (0.257)	-0.221 (3.385)	4.278 (3.095)	-0.012 (0.355)	0.056 (0.359)
Mothers Educ. (Bachelor's/Postgrad)	0.385 (0.718)	-0.338 (0.562)	-2.110 (2.641)	2.766* (1.526)	-0.199 (0.550)	0.768 (0.764)
Income Quintile	0.138 (0.405)	0.673 (0.733)	-0.096 (3.818)	0.408 (3.301)	-0.011 (0.468)	0.085 (0.707)
Fee Paying School	-0.201 (0.451)	0.073 (0.370)	0.342 (0.663)	-0.047 (0.239)	0.213 (0.463)	-0.038 (0.380)
DEIS School	1.665** (0.806)	0.735 (0.552)	2.334* (1.246)	0.205 (0.913)	-1.299* (0.765)	-0.102 (0.476)
Mixed School	0.045 (0.223)	0.041 (0.273)	-0.227 (2.581)	0.661 (2.080)	0.013 (0.244)	-0.061 (0.273)
Religious School	-0.049 (0.240)	-0.243 (0.671)	0.781 (3.206)	2.423 (3.130)	0.020 (0.238)	0.399 (0.717)
Constant	–	–	3.166 (9.052)	-11.219 (8.646)	–	–

Notes: The outcome is adjusted Leaving Certificate English points. The decomposition uses Wave 2 (age 13) predictors and is estimated separately for boys ($n = 1,300$; father present 1,178, father absent 122) and girls ($n = 1,289$; father present 1,138, father absent 151), using father-present coefficients as the reference group. Endowments capture differences in observed characteristics, coefficients capture differences in returns to those characteristics, and interaction captures their joint contribution. Positive values indicate components associated with higher scores among father-present students. Bootstrap standard errors are based on 1,000 replications. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

13 Appendix H. DFL Distributional Decomposition of Maths Gaps: Gender and Father Absence

While the main analysis employs Oaxaca-Blinder decompositions to examine mean differences, this approach cannot capture how gaps vary across the achievement distribution. This appendix reports the DFL results for both the gender comparison and the secondary father-absence extension in Maths. I complement the mean decompositions with DiNardo-Fortin-Lemieux (DiNardo et al., 1996) (DFL) decompositions, which allow examination of gender and father-absence gaps at specific quantiles (0.10, 0.25, 0.50, 0.75, 0.90), do not impose the linearity assumptions required by Oaxaca-Blinder, and reveal whether composition and structure effects differ for low-, medium-, and high-achieving students.

Choice of DFL over RIF

An alternative distributional decomposition approach is the recentered influence function (RIF) regression method of Firpo et al. (2009). The RIF for the τ -th quantile requires estimating $f_Y(q_\tau)$, the density of the outcome evaluated at the quantile, via kernel density estimation. Leaving Certificate Maths points are a bounded sum of discrete grade bands across six subjects ($H1=100$, $H2=88$, ...), creating a distribution with pronounced mass points where this density is not well defined in the continuous sense. Fortin et al. (2011) note explicitly that their distributional methods work best for “continuous and unbounded” outcome variables, and Rios-Avila (2020) demonstrate that the linear approximation underlying RIF regressions introduces specification error that compounds when the outcome has limited support. The DFL reweighting approach avoids this problem entirely: it operates on the empirical distribution function through propensity score reweighting and does not require density estimation at specific quantile points. For the same reason, DFL is better suited to the small father-absence cells ($N \approx 126$ – 150 per gender), where kernel density estimation would be particularly unreliable. I therefore adopt DFL as the primary distributional decomposition method.

Inference is based on 1,000 stratified bootstrap replications (resampling separately within each group to preserve group sizes), yielding pointwise 95% percentile confidence intervals for the total gap, composition effect, and structure effect at each quantile.

13.1 Decomposition of Gender Differences

The main DFL results for the gender gap in Maths are presented in the main text (Figure 3 and Table 3). This appendix expands on the estimation details and the density-based interpretation.

Following DiNardo et al. (1996), I construct a counterfactual distribution by reweighting girls’ Maths scores with propensity weights from a logistic model of gender on observables. Counterfactual quantiles are then computed directly as weighted quantiles of the girls’ outcome

distribution, which avoids extra simulation noise from random draws.

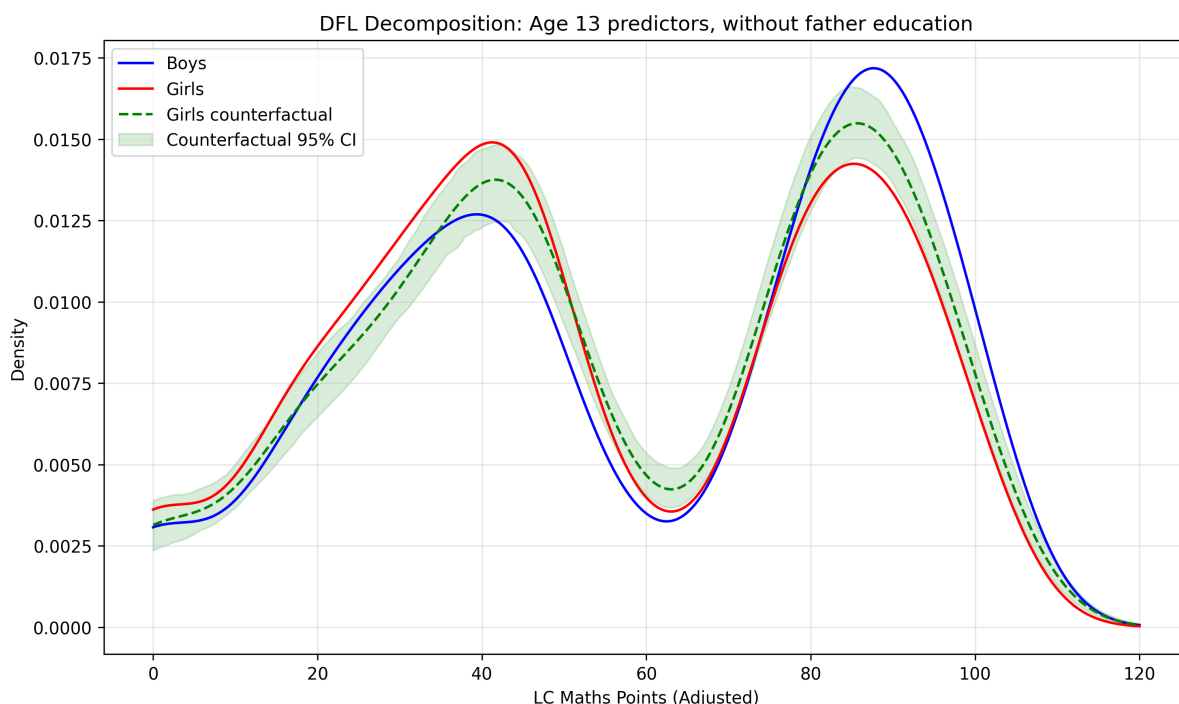


Figure 7: DFL density decomposition: age 13 predictors, without father's education. Maths score distributions for boys (blue), girls (red), and the counterfactual distribution (green dashed, with 95% bootstrap confidence band in shaded green). The counterfactual shows what girls' scores would look like if they had the same observable characteristics as boys. The bimodal shape reflects the Ordinary/Higher Level split in Leaving Certificate Maths. The composition effect (distance between red and green lines) is most visible in the middle of the distribution.

The distance between girls' observed scores (red) and the counterfactual (green) is the composition effect, while the distance between the counterfactual and boys' scores (blue) is the structure effect. In practice, both channels contribute, but composition becomes more visible around the middle of the distribution.

At lower quantiles (0.1–0.3), reweighting girls to boys' observables changes outcomes little and can even reduce predicted scores in some specifications. By contrast, at middle and upper quantiles, the composition effect becomes larger, indicating that observable characteristics explain more of the gap for middle and higher achievers.

Bootstrap intervals show that many quantile-specific composition estimates are imprecise, especially in the tails where effective cell sizes are smaller. At the 0.75 and 0.90 quantiles, composition is often statistically indistinguishable from zero, so structure effects and unobserved differences play a larger role there. Figure 7 also makes the bimodal Leaving Certificate Maths distribution explicit, consistent with the Ordinary/Higher level split, and shows the counterfactual lying between the male and female distributions.

13.2 Decomposition of Father Absence Effects

I then apply the same DFL framework to father-absence gaps, estimated separately by gender with predictors from age 9 (Wave 1) and age 13 (Wave 2). The counterfactual asks what father-present students would score if they had the observable profile of father-absent students.

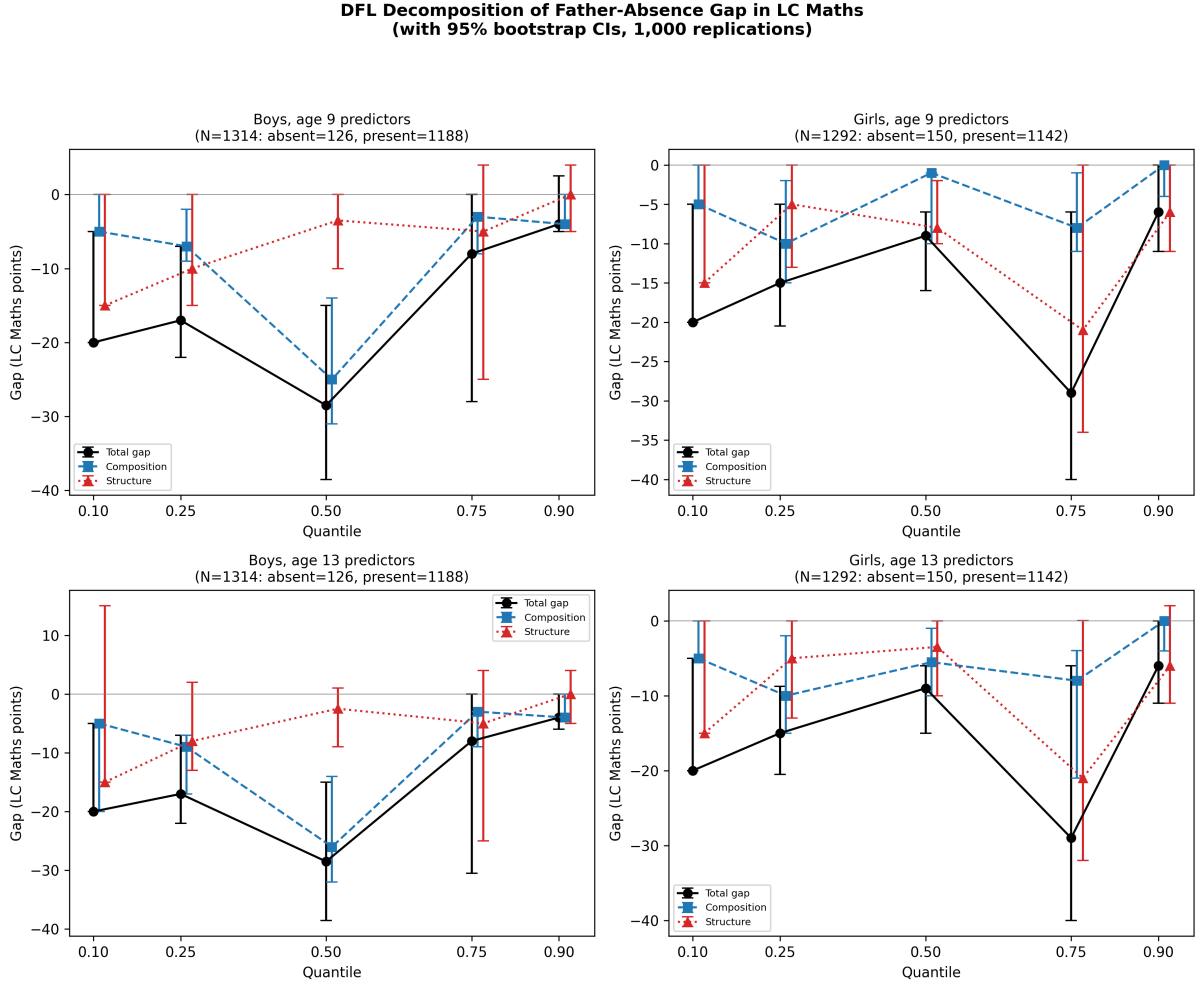


Figure 8: DFL decomposition of the father-absence gap in Leaving Certificate Maths across quantiles, by gender, with 95% bootstrap confidence intervals (1,000 stratified replications). Top row: age 9 predictors; bottom row: age 13 predictors; left column: boys; right column: girls. Black circles: total gap (absent – present). Blue squares: composition effect (counterfactual – present). Red triangles: structure effect (absent – counterfactual). Subtitles report cell sizes. The wider confidence intervals relative to the gender decomposition reflect the smaller father-absence subsamples ($N_{\text{absent}} \approx 126\text{--}150$ per gender).

Table 30 and Figure 8 show substantial father-absence penalties across the distribution. For boys (left panels), the total gap peaks at the median (-28.5 points, $SE = 6.3$), and composition explains 88–91% of that median gap across waves. At the lower tail, however, composition explains less, so structure effects account for more of the penalty.

For girls (right panels), the shape differs: the largest gap appears at $q_{0.75}$ (-29 points, $SE = 11.6\text{--}11.8$), and the composition share at the median is smaller, rising from 11% with

Table 30: DFL decomposition of adjusted Leaving Certificate Maths points by father absence

Specification	Component	$q_{0.10}$	$q_{0.25}$	$q_{0.50}$	$q_{0.75}$	$q_{0.90}$
Boys, W1	Total gap	-20.0 (6.9)	-17.0 (4.0)	-28.5 (6.3)	-8.0 (6.2)	-4.0 (2.2)
	Composition	-5.0 (1.8)	-7.0 (2.0)	-25.0 (5.4)	-3.0 (1.9)	-4.0 (1.4)
	Composition (%)	25%	41%	88%	38%	100%
Girls, W1	Total gap	-20.0 (5.9)	-15.0 (3.9)	-9.0 (2.7)	-29.0 (11.6)	-6.0 (4.0)
	Composition	-5.0 (2.1)	-10.0 (3.2)	-1.0 (2.7)	-8.0 (2.7)	0.0 (0.9)
	Composition (%)	25%	67%	11%	28%	0%
Boys, W2	Total gap	-20.0 (7.0)	-17.0 (3.9)	-28.5 (6.3)	-8.0 (6.8)	-4.0 (2.1)
	Composition	-5.0 (5.3)	-9.0 (2.2)	-26.0 (5.5)	-3.0 (2.1)	-4.0 (1.6)
	Composition (%)	25%	53%	91%	38%	100%
Girls, W2	Total gap	-20.0 (5.8)	-15.0 (3.9)	-9.0 (2.7)	-29.0 (11.8)	-6.0 (4.0)
	Composition	-5.0 (2.4)	-10.0 (3.4)	-5.5 (3.2)	-8.0 (4.2)	0.0 (1.2)
	Composition (%)	25%	67%	61%	28%	0%

Notes: Bootstrap standard errors are shown in parentheses and are based on 1,000 stratified replications. The total gap is father absent minus father present at each quantile. Composition is the counterfactual-minus-present component, and the structure component is the residual such that total equals composition plus structure. Father absence is defined as non-participation of the secondary caregiver in both the Wave 1 and Wave 2 parental surveys. The outcome is adjusted Leaving Certificate Maths points on a 0–100 scale.

age 9 predictors to 61% with age 13 predictors. Confidence bands are wider at $q_{0.75}$ because father-absent cells are small, so that quantile should be interpreted cautiously. For both genders, gaps narrow at the top of the distribution ($q_{0.90}$: 4–6 points).

Compared with the gender decomposition, the father-absence penalty is more broadly distributed across quantiles and is generally more composition-driven. This is consistent with the view that observed cognitive and family-background characteristics are more tightly linked to father-absence differences than to gender differences.

These distributional results complement the mean Oaxaca decompositions and show that father-absence associations are not confined to one part of the score distribution.

14 Appendix I. DFL Distributional Decomposition of Gender Gaps in English Achievement

The English DFL decomposition uses the same reweighting procedure and 1,000 stratified bootstrap replications. Table 4 and Figure 9 show that the English gender gap (3.1 points favouring girls) is concentrated near the lower tail and is small from the median upward. With age 13 predictors, the mean composition effect turns negative (-1.49 points), meaning boys hold stronger measured endowments while girls still score higher. The coefficient component therefore explains more than 100% of the mean gap at age 13, consistent with the subject reversal in the Oaxaca results.

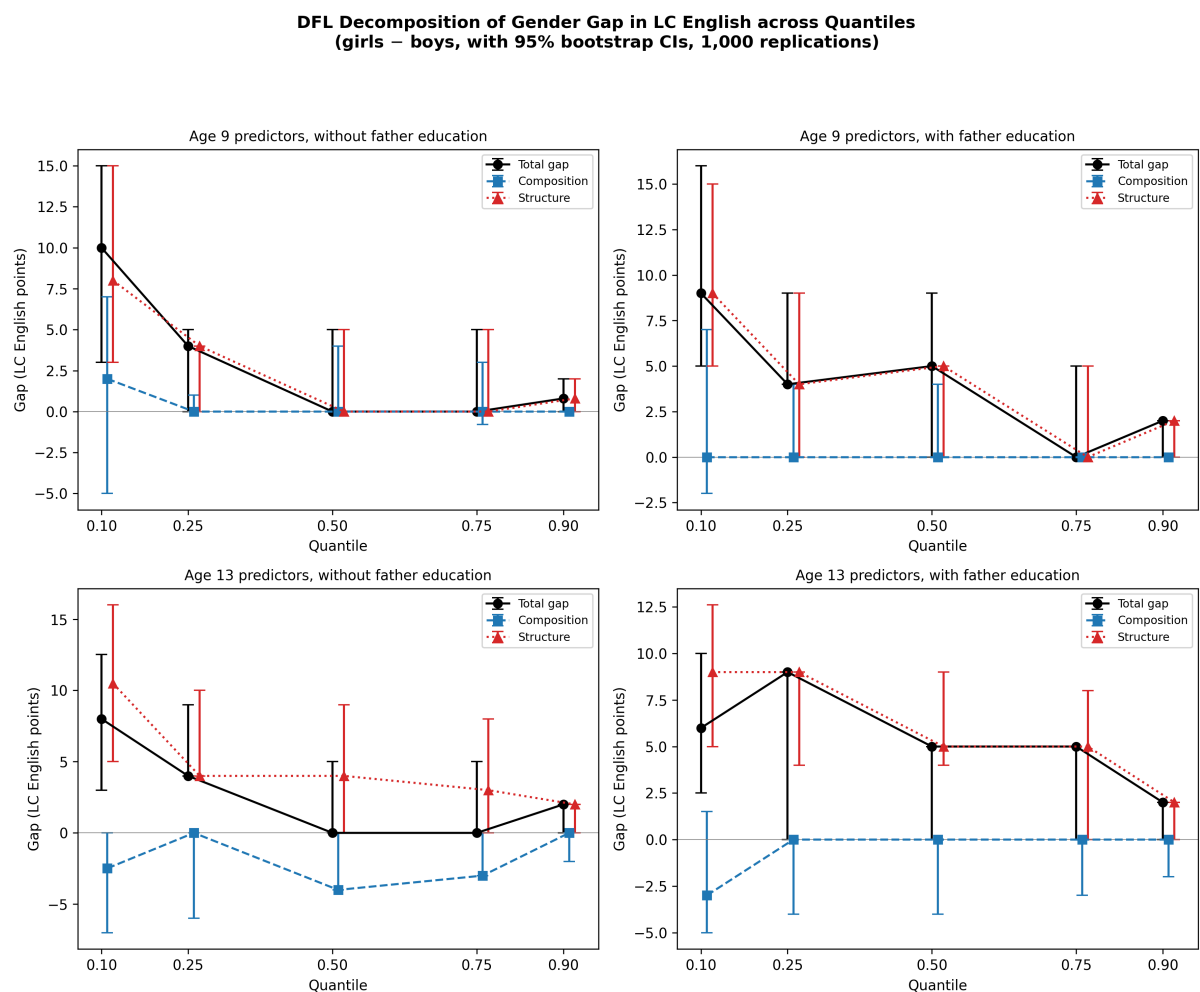


Figure 9: DFL decomposition of the gender gap in Leaving Certificate English across quantiles, with 95% bootstrap confidence intervals (1,000 stratified replications). Top row: age 9 predictors; bottom row: age 13 predictors; left column excludes father's education; right column includes it. The gender gap in English, which favours girls, is concentrated at the lower tail of the distribution.

15 Appendix J. DFL Distributional Decomposition of Father-Absence Gaps in English Achievement

Table 31 and Figure 10 present the English father-absence decomposition. For boys, the average penalty (5.7 points) is smaller than in Maths, and composition explains about two thirds of the gap across waves. For girls, the average penalty (4.2 points) has a distinctive Wave 2 profile in which composition exceeds the total gap, implying that measured-skill differences more than account for the observed difference while the structure component offsets part of it.

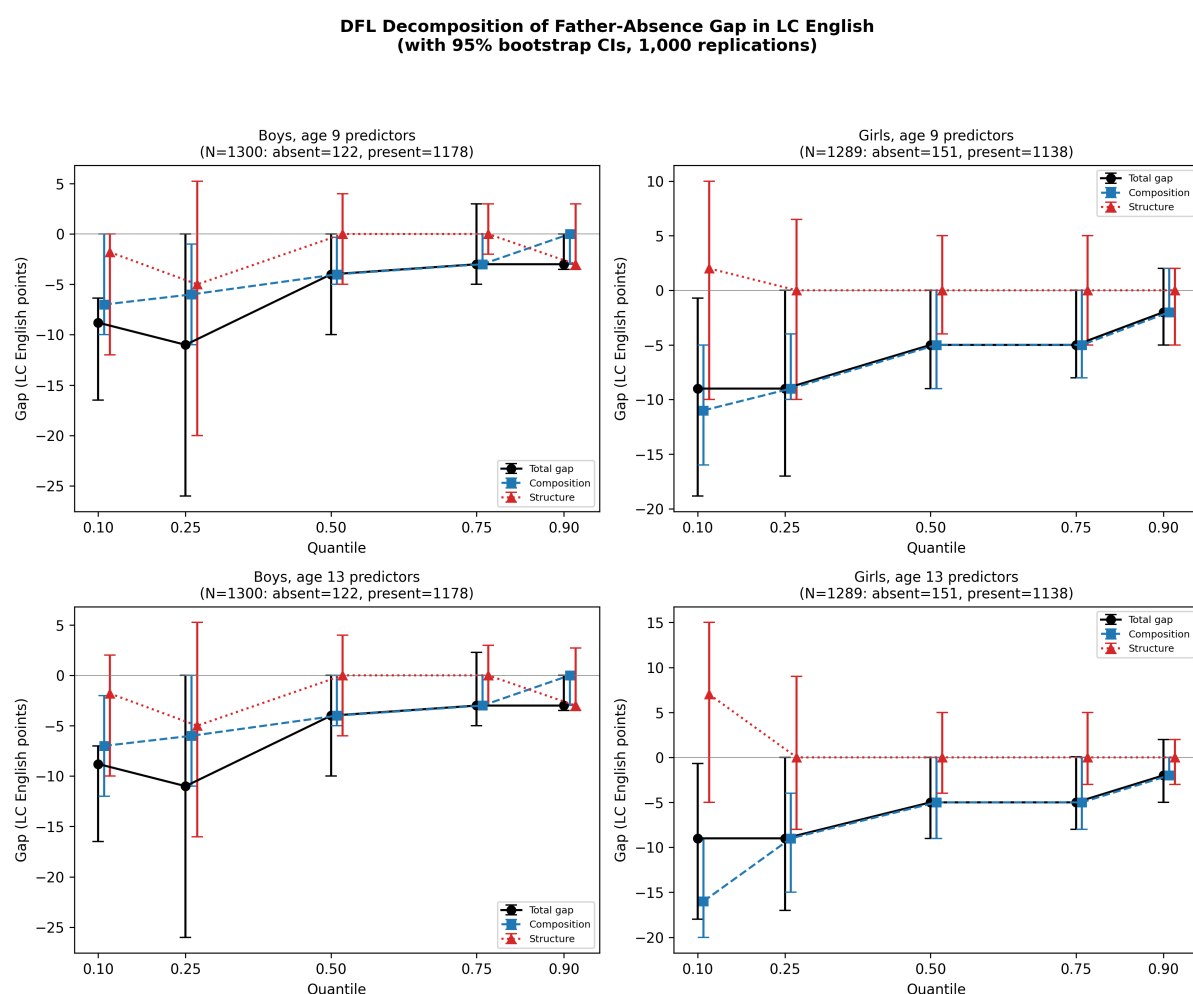


Figure 10: DFL decomposition of the father-absence gap in Leaving Certificate English across quantiles, by gender, with 95% bootstrap confidence intervals (1,000 stratified replications). Top row: age 9 predictors; bottom row: age 13 predictors; left column: boys; right column: girls.

Table 31: DFL decomposition of adjusted Leaving Certificate English points by father absence

Specification	Component	$q_{0.10}$	$q_{0.25}$	$q_{0.50}$	$q_{0.75}$	$q_{0.90}$
Boys, W1	Total gap	-8.8 (3.2)	-11.0 (6.3)	-4.0 (2.5)	-3.0 (1.9)	-3.0 (1.7)
	Composition	-7.0 (3.1)	-6.0 (3.3)	-4.0 (0.7)	-3.0 (0.9)	0.0 (1.1)
	Composition (%)	80%	55%	100%	100%	0%
Girls, W1	Total gap	-9.0 (4.8)	-9.0 (4.2)	-5.0 (2.3)	-5.0 (3.0)	-2.0 (2.1)
	Composition	-11.0 (3.5)	-9.0 (2.6)	-5.0 (1.8)	-5.0 (2.5)	-2.0 (1.0)
	Composition (%)	122%	100%	100%	100%	100%
Boys, W2	Total gap	-8.8 (3.0)	-11.0 (6.3)	-4.0 (2.5)	-3.0 (1.8)	-3.0 (1.6)
	Composition	-7.0 (2.6)	-6.0 (3.8)	-4.0 (1.5)	-3.0 (1.2)	0.0 (0.9)
	Composition (%)	80%	55%	100%	100%	0%
Girls, W2	Total gap	-9.0 (4.7)	-9.0 (4.4)	-5.0 (2.3)	-5.0 (3.1)	-2.0 (2.1)
	Composition	-16.0 (2.8)	-9.0 (2.7)	-5.0 (2.2)	-5.0 (2.4)	-2.0 (0.9)
	Composition (%)	178%	100%	100%	100%	100%

Notes: Bootstrap standard errors are shown in parentheses and are based on 1,000 stratified replications. The total gap is father absent minus father present at each quantile. Composition is the counterfactual-minus-present component, and the structure component is the residual such that total equals composition plus structure. Father absence is defined as non-participation of the secondary caregiver in both the Wave 1 and Wave 2 parental surveys. The outcome is adjusted Leaving Certificate English points.

16 Appendix K. School-Clustering Inference Sensitivity

This appendix reports a sensitivity exercise for unmodelled school clustering in the Oaxaca decompositions. Because school identifiers are unavailable in this analysis file, true cluster-robust or multilevel inference cannot be estimated directly. As a transparent second-best check, I inflate reported bootstrap standard errors by design-effect factors of 1.3 and 1.5, then re-evaluate headline significance for gender decompositions (Table 32) and father-absence decompositions (Table 33).

Table 32: Inference sensitivity to unmodelled school clustering in gender decompositions

Spec	Component	Estimate	SE	Baseline	×1.3	×1.5
M-W1-NF	Difference	-5.212	(0.980)	***	***	***
M-W1-NF	Endowments	-1.570	(0.610)	**	**	*
M-W1-NF	Coefficients	-4.215	(0.900)	***	***	***
M-W1-WF	Difference	-4.434	(1.020)	***	***	***
M-W1-WF	Endowments	-1.298	(0.656)	**	—	—
M-W1-WF	Coefficients	-3.641	(0.924)	***	***	***
M-W2-NF	Difference	-4.882	(0.998)	***	***	***
M-W2-NF	Endowments	-4.150	(0.742)	***	***	***
M-W2-NF	Coefficients	-1.236	(0.855)	—	—	—
M-W2-WF	Difference	-4.635	(1.150)	***	***	***
M-W2-WF	Endowments	-4.039	(0.822)	***	***	***
M-W2-WF	Coefficients	-1.128	(0.963)	—	—	—
E-W1-NF	Difference	3.079	(0.649)	***	***	***
E-W1-NF	Endowments	0.260	(0.398)	—	—	—
E-W1-NF	Coefficients	2.360	(0.618)	***	***	**
E-W1-WF	Difference	3.566	(0.677)	***	***	***
E-W1-WF	Endowments	0.456	(0.393)	—	—	—
E-W1-WF	Coefficients	2.821	(0.630)	***	***	***
E-W2-NF	Difference	2.906	(0.678)	***	***	***
E-W2-NF	Endowments	-1.588	(0.465)	***	***	**
E-W2-NF	Coefficients	4.548	(0.655)	***	***	***
E-W2-WF	Difference	3.135	(0.726)	***	***	***
E-W2-WF	Endowments	-1.123	(0.486)	**	*	—
E-W2-WF	Coefficients	4.396	(0.685)	***	***	***

Notes: Baseline columns use the reported bootstrap standard errors. Inflated columns multiply those standard errors by 1.3 and 1.5, respectively. This sensitivity exercise is motivated by GUI's school-clustered sampling and the absence of school identifiers in the analysis file. It does not replace cluster-robust inference; it shows how threshold-based significance changes under plausible design-effect multipliers. Specification codes are as follows: M/E = Maths/English; W1/W2 = predictor wave; NF/WF = no/with father's education. Significance codes follow the decomposition convention: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 33: Inference sensitivity to unmodelled school clustering in father-absence decompositions

Spec	Component	Estimate	SE	Baseline	×1.3	×1.5
M-FA-W1-B	Difference	13.564	(2.921)	***	***	***
M-FA-W1-B	Endowments	5.984	(2.502)	**	*	—
M-FA-W1-B	Coefficients	6.612	(2.655)	**	*	*
M-FA-W1-G	Difference	15.225	(2.415)	***	***	***
M-FA-W1-G	Endowments	3.122	(2.156)	—	—	—
M-FA-W1-G	Coefficients	7.456	(2.270)	***	**	**
M-FA-W2-B	Difference	13.564	(2.875)	***	***	***
M-FA-W2-B	Endowments	7.406	(2.456)	***	**	**
M-FA-W2-B	Coefficients	4.992	(2.482)	**	—	—
M-FA-W2-G	Difference	15.225	(2.520)	***	***	***
M-FA-W2-G	Endowments	7.405	(2.189)	***	***	**
M-FA-W2-G	Coefficients	6.351	(1.977)	***	**	**
E-FA-W1-B	Difference	5.673	(2.105)	***	**	*
E-FA-W1-B	Endowments	3.323	(1.869)	*	—	—
E-FA-W1-B	Coefficients	1.930	(1.890)	—	—	—
E-FA-W1-G	Difference	4.179	(1.707)	**	*	—
E-FA-W1-G	Endowments	2.980	(1.554)	*	—	—
E-FA-W1-G	Coefficients	0.742	(1.572)	—	—	—
E-FA-W2-B	Difference	5.673	(2.036)	***	**	*
E-FA-W2-B	Endowments	4.112	(1.864)	**	*	—
E-FA-W2-B	Coefficients	1.975	(1.728)	—	—	—
E-FA-W2-G	Difference	4.179	(1.672)	**	*	*
E-FA-W2-G	Endowments	4.376	(1.469)	***	**	**
E-FA-W2-G	Coefficients	-0.684	(1.492)	—	—	—

Notes: Baseline columns use the reported bootstrap standard errors. Inflated columns multiply those standard errors by 1.3 and 1.5, respectively. This sensitivity exercise is motivated by GUI's school-clustered sampling and the absence of school identifiers in the analysis file. It does not replace cluster-robust inference; it shows how threshold-based significance changes under plausible design-effect multipliers. Specification codes are as follows: M/E = Maths/English; FA = father-absence decomposition; W1/W2 = predictor wave; B/G = boys/girls. Significance codes follow the decomposition convention: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

17 Appendix L. Level-Selection Sensitivity in Leaving Certificate Maths

Because Leaving Certificate Maths points pool Higher and Ordinary levels, this appendix tests how sensitive the main gender decompositions are to that mixture. Figure 11 shows the bimodal distribution and the data-driven antimode split at 65.0 points used to define lower- and upper-support samples.⁶ I then re-estimate the four main gender Oaxaca specifications (Wave 1/2, with/without father's education) within each support band (Table 34).

Interpretation: the restricted-support results show that the pooled Maths gap is concentrated in the upper support. Within the lower support, the gap is essentially zero in all four specifications

⁶The visible mass at 0 points is retained as a valid floor score under the adjusted Leaving Certificate Maths mapping. These observations therefore remain in the analytical sample.

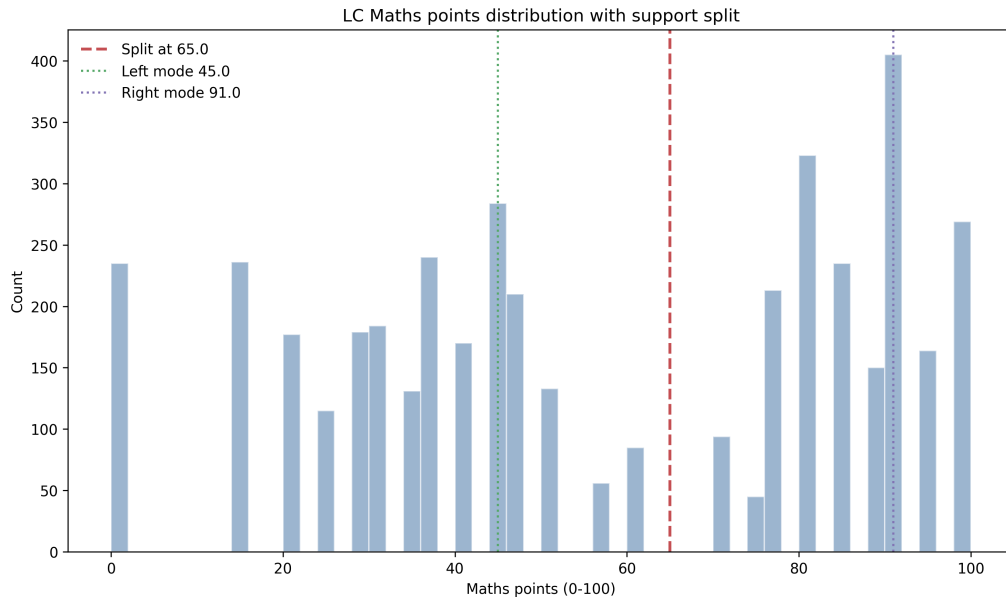


Figure 11: LC Maths points distribution with data-driven support split at the empirical antimode (65.0 points). The vertical dashed line marks the split used for lower- versus upper-support sensitivity samples in Appendix 17. Dotted lines indicate the two modal regions.

and is slightly positive in three of them, so there is little to decompose. Within the upper support, the gap is modest in size, around 1.2 to 1.4 points, and remains coefficient-dominant at both Wave 1 and Wave 2. The Wave 1 to Wave 2 shift toward endowment dominance in the full sample therefore reflects largely differential placement into the upper support. A within-support shift of the same magnitude does not appear in these results. This is consistent with the extensive-intensive decomposition in Appendix 19, where the main timing shift is concentrated in level selection. Points conditional on Higher Level participation account for a much smaller part of the shift.

Table 34: Support-restriction sensitivity of the Leaving Certificate Maths gender decomposition

Specification	Sample	Difference	Endowments	Coefficients	Endowment share
W1, no father ed.	Full sample	-5.212	-1.570	-4.215	30.1%
	Lower support	0.066	0.554	-0.327	n/a
	Upper support	-1.257	-0.135	-1.254	10.7%
W1, with father ed.	Full sample	-4.434	-1.298	-3.641	29.3%
	Lower support	0.911	0.639	0.231	n/a
	Upper support	-1.243	-0.128	-1.279	10.3%
W2, no father ed.	Full sample	-4.882	-4.150	-1.236	85.0%
	Lower support	0.280	-0.623	1.076	n/a
	Upper support	-1.426	-0.212	-1.241	14.9%
W2, with father ed.	Full sample	-4.635	-4.039	-1.128	87.1%
	Lower support	0.435	-0.373	1.252	n/a
	Upper support	-1.236	-0.288	-1.146	23.3%

Notes: The sample is split at the antimode of the observed bimodal score distribution (65.0 points); Oaxaca components are then re-estimated separately in the lower- and upper-support subsamples. Endowment shares are not reported for the lower-support sample because the total gap there is close to zero, so the share is not meaningful. The exercise is intended as a sensitivity check, not as a level-specific identification strategy. The split is data-driven and is intended to assess whether the Wave 1 to Wave 2 composition shift is mechanically driven by differences in level mix in the pooled points outcome.

18 Appendix M. Attrition Robustness via Lee Bounds

This appendix reports Lee (2009) monotonic-selection bounds for the father-absence gap in Leaving Certificate Maths points. The treatment group is father-absent status (Waves 1 and 2 non-response), and outcomes are observed only for cases with valid Wave 4 Leaving Certificate reports. Because selection is lower in the father-absent group, the Lee procedure trims the selected father-present distribution to match that selection rate. Table 35 reports the bounds.

Table 35: Lee bounds for the father-absence gap in Leaving Certificate Maths points

Sample	Naive diff.	Lower bound	Upper bound	Sel. absent	Sel. present	Trim share
Overall	-18.303	-30.549	-5.730	57.8%	73.1%	20.9%
Girls	-16.460	-27.857	-4.075	58.0%	73.3%	20.9%
Boys	-19.689	-32.791	-7.125	57.6%	72.9%	21.0%

Notes: Lee (2009) monotonic-selection bounds for the father-absence gap in Leaving Certificate Maths points. Treatment is father-absent status, defined as non-response in both Wave 1 and Wave 2. The outcome is Leaving Certificate Maths points. Bounds are reported for father absent minus father present, so negative values indicate a penalty associated with father absence. When selection is lower in the father-absent group, the selected father-present outcome distribution is trimmed to match the father-absent selection rate under the monotonic-selection assumption. This yields worst-case bounds for attrition bias from differential selection into observed outcomes.

Interpretation: bounds remain negative overall and by gender, indicating that the father-absence penalty does not disappear under monotonic-selection correction, even though magnitudes are interval-valued rather than point-identified.

19 Appendix N. Additional Defence Robustness Checks

This appendix reports three targeted checks. First, a harmonised Wave 1 versus Wave 2 timing test retains only conceptually matched covariates across waves; the endowment-share shift remains strong in Maths and reverses in English. Second, the level-mix concern is split into an extensive margin (probability of upper-support placement) and an intensive margin (points conditional on upper support), reported separately to clarify where composition effects are concentrated. Third, father-absence decompositions are re-estimated under alternative definitions using Wave 3 partner and SCG-participation information to separate structural absence from resident disengagement.

Why the timing shift is not just changing covariates. The baseline decompositions use wave-specific predictor sets because the GUI instruments change between ages 9 and 13. A natural objection is that the coefficient-to-endowment shift may come from richer or better-scaled Wave 2 measurement. The harmonised specification addresses this by restricting both waves to covariates that share the same conceptual content:

- Covariates available at both waves (retained in harmonised specification): one numeracy measure, one reading/verbal reasoning measure, four SDQ subscales (emotional, conduct, hyperactivity, peer problems), primary caregiver education (two dummies), equivalised household income quintile, mixed-school indicator (the co-educational-school measure available at both waves), father-education-missing indicator, and a grading-system dummy.
- Covariates that differ across waves: the cognitive instruments themselves change (BAS-derived scores at Wave 1; Drumcondra Numerical Ability and Verbal Reasoning at Wave 2). These are conceptually matched (numeracy and verbal/reading) but not identical tests.
- Covariates available only at Wave 2 (excluded from harmonised specification): fee-paying school status, DEIS status, and secondary caregiver education.

Tables 36 and 37 report the result. With these harmonised covariate sets, the endowment-share shift is 51 percentage points in Maths (no father-education control, $p = 0.002$) and 62 percentage points (with father-education control, $p = 0.048$). In English, the shift reverses direction by 35–53 percentage points ($p = 0.013$ and $p = 0.017$). The corresponding changes in the endowment component in levels are also negative and precisely estimated across all four specifications. One Maths specification, the Wave 2 harmonised model with observed father's education, produces an endowment share slightly above 100%. Substantively, that means the measured endowment advantage is larger than the observed total gap and the coefficients component partly offsets it. This does not overturn the timing claim. Both harmonised Maths specifications still move sharply toward endowment dominance, and the endowment component

in levels rises by nearly identical amounts in the two samples (-2.79 and -2.75 points). The timing pattern therefore does not depend on Wave 2-specific school controls or the availability of additional predictor variables; it survives when the covariate structure is held as constant as the data allow.

Table 36: Wave-comparability check: endowment shares and shifts

Specification	W1 (%)	W2 (%)	Shift	SE	<i>p</i>
Maths, no father	38.98	89.94	50.96	16.44	0.002
Maths, with father	39.64	101.61	61.97	31.37	0.048
English, no father	-1.03	-54.38	-53.35	21.51	0.013
English, with father	6.82	-29.03	-35.86	15.03	0.017

Notes: Endowment shares are reported for Wave 1 and Wave 2 using harmonised covariate sets. *Shift* is the change in the endowment share (Wave 2 minus Wave 1). Standard errors and *p*-values are based on bootstrap inference.

Table 37: Wave-comparability check: differences in endowment components

Specification	Diff. endow.	SE	<i>p</i>
Maths, no father	-2.79	0.65	0.000
Maths, with father	-2.75	0.73	0.000
English, no father	-1.69	0.41	0.000
English, with father	-1.43	0.45	0.000

Notes: *Diff. endow.* reports the change in the endowment component in levels (Wave 2 minus Wave 1) using harmonised covariate sets. Standard errors and *p*-values are based on bootstrap inference.

Why the timing shift is not just level mix. A further concern is that the stronger Wave 2 composition result in Maths may simply reflect changing mass in the upper part of the bimodal score distribution. On that reading, the result would say less about a broader shift in the determinants of achievement. This concern is especially relevant in Ireland, where bonus points for Higher Level Maths have changed participation incentives and the profile of students taking the subject (O’Meara et al., 2020). To assess this, the sample is split at the antimode (65.0 points), and the decomposition is estimated separately for the extensive margin of upper-support placement and the intensive margin of points conditional on upper support. This complements the support-restriction sensitivity in Appendix 17, which shows the same concern through pooled-score Oaxaca decompositions within each support band. Tables 38 and 39 show that the large Wave 2 composition share is concentrated in the extensive margin, where endowments account for about 89% of the shift, whereas the intensive-margin gap conditional on upper support remains much smaller and only about 18–20% endowment-led. In other words, the intensive-margin gap stays coefficient-dominant at both waves. The main timing shift is therefore concentrated in level selection and not in within-level points differences alone.

Table 38: Two-part analysis: extensive margin (upper-support probability)

Specification	Wave	Father ed.	<i>N</i>	Difference	Endow.	Coef.
	W1	No	3,690	-0.086	-0.034	-0.063
	W2	No	3,617	-0.078	-0.070	-0.021
	W1	Yes	3,241	-0.081	-0.029	-0.059
	W2	Yes	2,940	-0.075	-0.067	-0.020

Notes: Outcome is $P(\text{upper support})$ using a split at 65.0. Reported components are Oaxaca–Blinder decompositions.

Table 39: Two-part analysis: intensive margin (conditional points)

Specification	Wave	Father ed.	<i>N</i>	Difference	Endow.	Coef.
	W1	No	1,581	-1.257	0.018	-1.342
	W2	No	1,576	-1.380	-0.252	-1.265
	W1	Yes	1,454	-1.243	0.011	-1.358
	W2	Yes	1,363	-1.254	-0.256	-1.220

Notes: Outcome is points conditional on being in the upper-support region. Reported components are Oaxaca–Blinder decompositions.

Why the father-absence result is not driven by the baseline proxy. The main analysis defines father absence using consistent non-response to the secondary-caregiver questionnaire in Waves 1 and 2. A natural concern is that this proxy may combine structurally absent fathers with resident but disengaged fathers. Table 40 addresses this by re-estimating the decomposition under alternative definitions based on Wave 3 partner status and SCG participation. The broad timing pattern remains: penalties are larger under structural absence than under resident disengagement, and Wave 2 endowment shares are generally higher than their Wave 1 counterparts. One resident-disengagement specification, girls at Wave 2, produces an endowment share of 121%. Here the measured endowment disadvantage exceeds the observed total gap and the coefficient component turns slightly negative, so the coefficients structure partly offsets the penalty. That is unusual but still consistent with the broader appendix pattern: measured disadvantages are most severe under structural absence, while resident disengagement yields smaller penalties and weaker coefficient amplification.

Table 40: Robustness to alternative definitions of father absence

Definition	Sample	<i>N</i>	Difference	Endow.	Coef.	Share (%)
Baseline (W1+W2 non-response)	W1 All	3,153	14.433	5.182	6.322	35.906
	W1 Boys	1,570	13.172	5.637	6.017	42.795
	W1 Girls	1,583	14.876	3.750	6.482	25.207
	W2 All	3,164	15.333	8.502	5.882	55.446
	W2 Boys	1,564	14.685	6.894	5.714	46.948
	W2 Girls	1,600	15.412	8.314	6.077	53.942
Structural (no partner, W3)	W1 All	2,532	13.633	6.082	7.085	44.615
	W1 Boys	1,264	12.535	5.604	6.860	44.709
	W1 Girls	1,268	13.830	4.695	7.252	33.948
	W2 All	2,652	13.450	7.741	5.012	57.558
	W2 Boys	1,314	12.640	7.641	5.336	60.455
	W2 Girls	1,338	13.467	6.687	4.944	49.656
Resident disengaged (W3)	W1 All	2,634	5.444	3.025	2.249	55.571
	W1 Boys	1,326	5.479	2.143	2.851	39.114
	W1 Girls	1,308	5.032	4.038	1.466	80.246
	W2 All	2,756	5.230	4.086	0.998	78.123
	W2 Boys	1,376	5.547	2.620	2.654	47.223
	W2 Girls	1,380	4.586	5.549	-0.377	121.011

Notes: Entries report Oaxaca–Blinder decompositions of the father-absence gap under alternative definitions. The baseline definition uses consistent non-response in Waves 1 and 2. The structural definition uses no resident partner at Wave 3. The resident-disengagement definition uses a resident partner without SCG completion at Wave 3.

20 Appendix O. Attrition Balance by Gender

Table 41 compares Wave 1 baseline covariates for children retained in the baseline-overlap analytical sample ($N = 4,213$) versus those lost to attrition ($N = 1,681$), separately by gender and pooled. Standardised differences exceeding $|0.10|$ indicate non-trivial imbalance; values exceeding $|0.25|$ are conventionally large.

Retained children score 0.3–0.4 SD higher on Wave 1 cognitive tests and come from households with higher income and more educated primary caregivers. They also display fewer externalising behaviours (lower SDQ conduct and hyperactivity scores), are slightly less likely to have missing father education, and look almost identical to lost cases on Wave 1 co-educational school exposure. Attrition rates themselves are effectively equal by gender (2,050/2,870 boys retained, 71.4%; 2,163/3,024 girls retained, 71.5%). The imbalance pattern is broadly similar in direction for boys and girls, though magnitudes on some behavioural and parental education variables are moderately larger for girls. The retained-vs.-lost imbalance on father-education missingness also implies that the with-father-education decomposition samples are further selected on parental engagement, an additional reason to read those specifications as conditional on a more advantaged subpopulation. These imbalances motivate the IPW reweighting reported in Table 12 and the Lee bounds in Appendix 18.

Table 41: Attrition balance on Wave 1 covariates, by gender

Variable	Mean (retained)	Mean (lost)	Std. diff.
<i>Boys</i> ($N_{\text{retained}} = 2,050$; $N_{\text{lost}} = 820$)			
Cognitive Maths (W1, log)	-0.377	-0.707	+0.355
Cognitive Reading (W1, log)	0.353	-0.069	+0.431
SDQ Emotional (W1)	1.811	1.908	-0.052
SDQ Conduct (W1)	1.190	1.460	-0.189
SDQ Hyperactivity (W1)	3.025	3.587	-0.228
SDQ Peer Problems (W1)	1.068	1.202	-0.092
PCG Educ: Secondary (W1)	0.557	0.571	-0.028
PCG Educ: Degree+ (W1)	0.336	0.244	+0.200
Income Quintile (W1)	3.553	3.239	+0.237
Co-educational School (W1)	0.697	0.711	-0.030
Father Educ Missing (W1)	0.109	0.165	-0.168
<i>Girls</i> ($N_{\text{retained}} = 2,163$; $N_{\text{lost}} = 861$)			
Cognitive Maths (W1, log)	-0.556	-0.830	+0.319
Cognitive Reading (W1, log)	0.355	0.021	+0.357
SDQ Emotional (W1)	2.049	2.252	-0.102
SDQ Conduct (W1)	1.037	1.345	-0.231
SDQ Hyperactivity (W1)	2.421	3.030	-0.273
SDQ Peer Problems (W1)	1.071	1.167	-0.069
PCG Educ: Secondary (W1)	0.561	0.585	-0.050
PCG Educ: Degree+ (W1)	0.310	0.193	+0.264
Income Quintile (W1)	3.463	3.186	+0.207
Co-educational School (W1)	0.824	0.804	+0.051
Father Educ Missing (W1)	0.130	0.201	-0.198
<i>Pooled</i> ($N_{\text{retained}} = 4,213$; $N_{\text{lost}} = 1,681$)			
Cognitive Maths (W1, log)	-0.469	-0.770	+0.336
Cognitive Reading (W1, log)	0.354	-0.023	+0.394
SDQ Emotional (W1)	1.933	2.085	-0.078
SDQ Conduct (W1)	1.111	1.401	-0.209
SDQ Hyperactivity (W1)	2.715	3.302	-0.248
SDQ Peer Problems (W1)	1.069	1.184	-0.081
PCG Educ: Secondary (W1)	0.559	0.578	-0.039
PCG Educ: Degree+ (W1)	0.323	0.218	+0.232
Income Quintile (W1)	3.507	3.212	+0.221
Co-educational School (W1)	0.761	0.759	+0.006
Father Educ Missing (W1)	0.120	0.183	-0.184

Notes: The baseline is the broader pre-analytical sample ($N = 5,894$). Retained observations here are the subset of the decomposition analytical sample that also appear in the baseline-overlap file used for attrition comparison ($N = 4,213$). This is narrower than the full valid-outcome decomposition sample of $N = 4,333$. The standardised difference is defined as $(\bar{x}_{\text{retained}} - \bar{x}_{\text{lost}})/s_{\text{pooled}}$. Absolute values above 0.10 indicate non-trivial imbalance, and values above 0.25 are conventionally large.

21 Appendix P. Leaving Certificate Maths Reporting Diagnostics

A concern with self-reported examination outcomes is that respondents may round, heap, or misreport their scores. Figure 12 plots the distribution of Leaving Certificate Maths points in the analytical sample, first by gender and then by grading system.

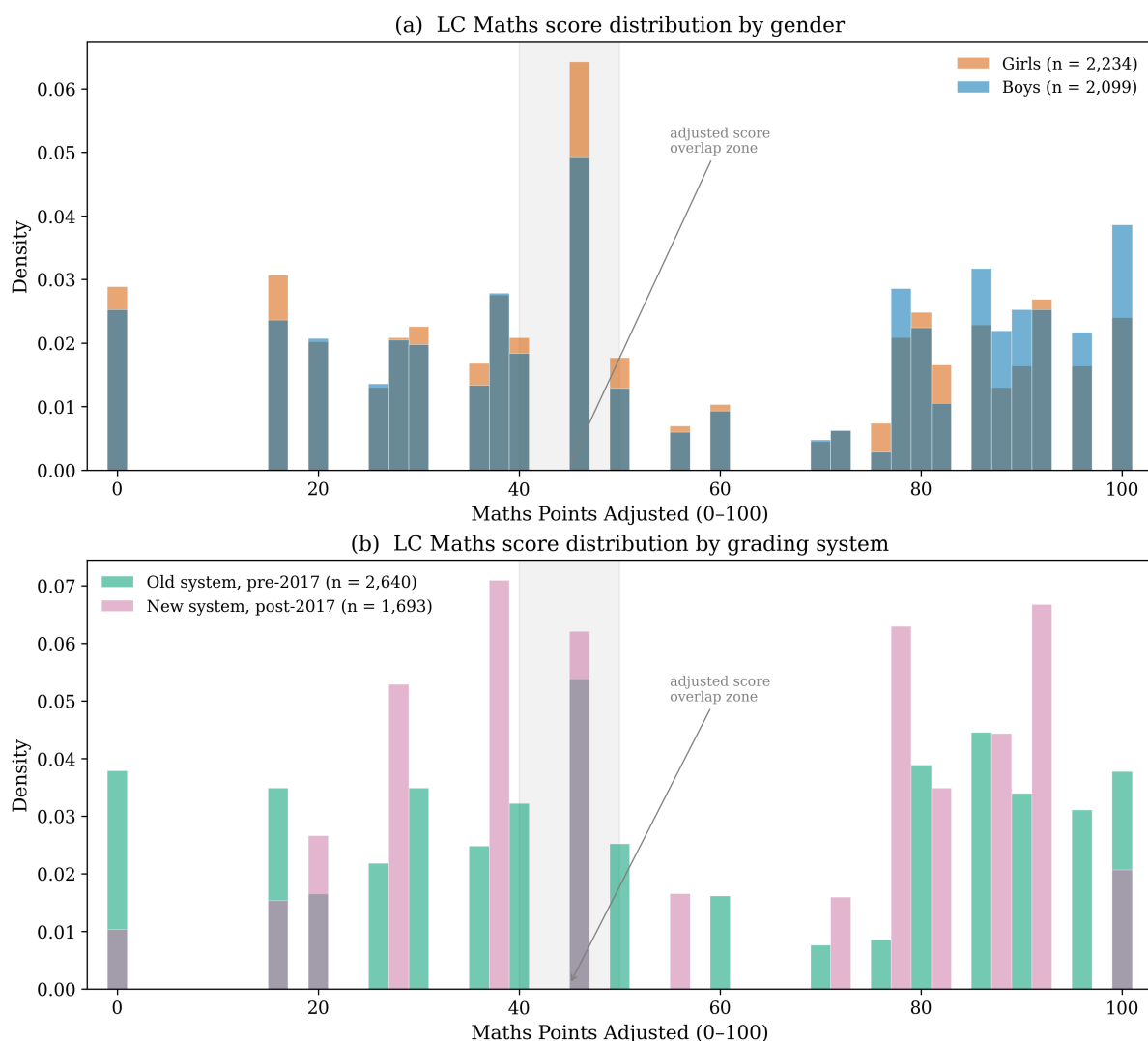


Figure 12: Distribution of self-reported Leaving Certificate Maths points ($N = 4,333$). Panel (a) reports the analytical sample by gender; panel (b) reports the same outcome by grading system. The distribution exhibits 26 discrete values corresponding to grade boundaries under the old and new Leaving Certificate grading systems, producing a clear bimodal structure. The absence of inter-boundary heaping suggests that respondents report actual grades rather than approximate point totals.

The distribution shows 26 discrete values aligned with official grade boundaries, with no sign of a continuous or heaped pattern. For the reporting diagnostic, the key feature is shape: there is no visible inter-boundary heaping at round numbers that would suggest free-form point

estimation. Instead, respondents appear to report letter grades that map mechanically into points. Panel (b) shows that this discretisation is present under both the pre-2017 and post-2017 grading systems, with different interior mass points because the reform changed the underlying grade-to-points grid. The prominent spike around 45 points is therefore expected under the adjusted score mapping: it is one of the densest overlap regions in the old and new grade grids after bonus-point subtraction, not evidence of rounding. This does not rule out systematic level-selection bias (addressed in Appendix 17), but it does indicate that any remaining misreporting operates primarily at the grade level and does not take the form of free-form point reporting.